Procuring Hunting Effort: Structural Estimation and Optimal Contract Design

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March 17, 2026

Abstract

This paper analyzes the procurement of wildlife damage control under informational frictions and model uncertainty, focusing on deer management in Hokkaido, Japan. I develop a structural state-space model to estimate hunter responses to two prevalent incentive schemes: per-diem "dispatch" payments (input-based) and per-animal "bounty" payments (output-based). Using administrative records from 2022–2024, the empirical estimation reveals that while both schemes increase harvests, they differ in their impact on damage reduction: bounty-based incentives exhibit a "capture-damage gap" where increased harvests are positively correlated with realized damages. I then design optimal and robust contracts by accounting for model ambiguity regarding hunters' behavioral margins and ecological linkages. The results show that while a standard optimal contract favors cost-effective per-diem payments, a max-min robust contract shifts toward bounty-based incentives. This shift occurs because bounties activate multiple behavioral channels, serving as a hedge that ensures stable performance across a range of model specifications. These findings provide an economic rationale for the coexistence of bounty schemes in practice as a tool for ensuring robustness against uncertainty.

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1 Introduction

Wildlife damage—such as deer-induced crop losses—is widely viewed as a domain where government plays a central role. Because animals move across administrative boundaries, local actions can affect outcomes beyond the jurisdiction, providing a rationale for coordinated management (Bhat et al., 1996). A common focus in the economics discussion has been *ex post* policy on the victim side, especially public compensation for realized losses and the incentive problems such programs create for prevention and reporting (Rollins and Briggs, III, 1996; Rondeau and Bulte, 2007). But effective control is, in the end, a *supply-side* problem: damages fall only if agencies can procure sufficient, well-targeted hunting effort. Doing so requires contracting with hunters and contractors under limited observability, noisy and imperfect performance measures, and scope for strategic behavior. This is precisely where contract theory offers a practical intervention as a toolkit for policy design—clarifying what should be measured, how performance should be rewarded, and how programs can be made robust to manipulation.

Procuring such effort is complicated by informational and behavioral frictions that vary across contract types. Under input-based schemes, such as per-diem wages, agencies face moral hazard and "budget consumption," where hunters lack financial motivation to maximize capture efficiency. Conversely, output-based incentives, such as per-animal bounties, can lead to a "capture-damage gap": rational hunters may prioritize easy-to-catch animals in remote high-density habitats rather than focusing on agricultural fringes where capture is difficult but damage reduction is most vital (Hone, 2007; Kaji et al., 2013). This paper brings a contract-theoretic lens to this procurement problem, shifting the policy focus from simple population reduction to targeted damage mitigation. For the deer management problem, I develop an integrated framework from estimating a structural model to designing optimal compensation schemes that are robust to model ambiguity.

Deer management in Hokkaido provides a natural empirical setting for studying contract design. Deer-induced agricultural damage is a persistent issue, prompting policymakers to evaluate the effectiveness of current contracting methods with hunters. In practice, the management regime consists of two primary schemes: a dispatch-based payment (Plan A) and a per-animal bounty (Plan B). Plan A compensates hunters via per-diem payments for organized dispatch operations, focusing on input. In contrast, Plan B provides fixed payments for each harvested deer, focusing on output. Currently, these two instruments coexist, with bounty-based payments accounting for the majority of total expenditure. However, there is limited evidence on the optimality of this policy mix. Consequently, there is a need for payment rules that maintain performance under uncertainty regarding hunters' behavioral responses and ecological conditions.

This paper uses administrative records compiled by the Hokkaido prefectural government: region-level hunting implementation reports for fiscal years 2022–2024. These reports contain, for each region and contract type, the number of animals harvested, total payments, and operational inputs such as labor expenditures under the cost-plus scheme and the number of dispatches. Importantly, the same reports also record the policy-relevant outcome measure: region-level monetary damages attributed to deer.

I develop a behavioral model of hunting activity under the two contract types. A representative hunter chooses dispatch frequency and effort intensity, and these choices jointly determine harvests. The model delivers reduced-form predictions for log harvests under each contract type, which form the backbone of the estimating equations. Estimation proceeds by maximum likelihood in a state-space model in which local deer abundance is a latent state evolving over time. A key empirical challenge is endogeneity among harvests, damages, and incentives. To address this concern, I exploit cross-regional variation by using differences relative to neighboring regions to separately identify the relevant behavioral and ecological linkages.

Our empirical estimation yields three primary findings. First, regardless of the regime, both per-diem payments and per-animal bounties significantly increase total harvests. Second, while these additional harvests translate into damage reduction under the dispatch-based scheme, we find a perverse effect under the bounty-based scheme: increased catches are associated with higher damages. This result is consistent with the hypothesis that output-based incentives shift hunting behavior in a manner that decouples measured harvests from actual damage mitigation. Third, the structural estimates reveal that while effort has a substantial marginal effect on harvests, its marginal cost escalates more rapidly than the cost of dispatch. Consequently, the relative efficiency of inducing direct dispatch versus incentivizing broader effort remains a non-trivial policy trade-off.

We evaluate these trade-offs by comparing optimal contracts with robust contracts that account for model ambiguity. While the optimal contract relies on per-diem payments to utilize the cost-effectiveness of dispatch, the robust contract incorporates bounty-based incentives. This shift occurs because bounties activate both dispatch and effort margins. By loading on multiple behavioral channels, the bounty instrument provides more stable performance across different model specifications than a dispatch-only contract. These results suggest that bounty-based schemes in practice can be interpreted as a tool for ensuring robustness against uncertainty in hunter behavior and ecological

responses.

Literature review. This paper relates to the classical comparison between cost-plus contracts, which reimburse realized costs, and fixed-price contracts, which pay a predetermined amount for performance (Laffont and Tirole, 1993; Bajari and Tadelis, 2001; Rogerson, 2003; Lafontaine and Slade, 2012). A growing empirical literature studies the incidence and performance of these contract forms in realistic procurement settings using administrative and transaction-level data (Corts and Singh, 2004; Gagnepain et al., 2013; Carril and Duggan, 2020). Much of this work, however, focuses on which contract form is selected and how it performs along proximate procurement metrics (e.g., costs, delivery, or renegotiation), rather than using the data to redesign incentives and solve for an optimal contract within a feasible policy class.

This paper also relates to the rapidly expanding literature on robust contracting and robust mechanism design, much of which has been developed at the interface of economic theory and computation. Recent work formalizes robustness as performance guarantees under model misspecification and limited knowledge of the underlying environment, and derives sharp characterizations of simple contractual forms. A prominent theme is that robustness considerations can rationalize the prevalence of simple contracts observed in practice (e.g., Carroll (2015); see also Dütting et al. (2018); Peng and Tang (2024)). Related contributions study how robust contractual prescriptions can be informed by data and behavioral evidence (Antic and Georgiadis, 2024), and how robustness considerations reshape procurement design more broadly (Mishra et al., 2025).

A practical obstacle for empirical work is that robustness is not a single prescription: different criteria can, in principle, select different functional forms and thus lead to different policy recommendations. Prominent benchmarks include max–min (worst-case) performance, regret-based evaluations in the spirit of Bergemann and Schlag (2008), and ratio-regret objectives as in Hartline and Roughgarden (2014). Moreover, much of the robust-contracting literature is developed in stylized environments, and there is limited evidence on how much structure an empirical researcher must impose—and which components must be estimated rather than calibrated—in order to compute robust contracts in applied settings. In particular, it is often unclear how to align a robustness objective with an empirically implementable model and identification strategy, while preserving a tractable contract space that can be optimized in practice. This paper addresses these

implementation issues directly. I develop an empirically disciplined robust-contracting framework in which the ambiguity set is anchored to estimable objects, the counterfactual mapping from incentives to outputs and outcomes is fully specified, and the robust optimization problem remains computationally tractable under the administrative and institutional constraints of the policy environment.

2 Background Information and Data

2.1 Deer Management and Payment Schemes in Hokkaido

Hokkaido in Japan provides a unique setting for studying wildlife management due to its island geography and the significant economic impact of the Hokkaido sika deer (*Ezo deer*). In FY2022, deer accounted for approximately 82% of the total wildlife-related agricultural damages in the prefecture, with losses heavily concentrated in forage and pasture crops.[1] Since forage is a critical input for Hokkaido's dairy and livestock industry, these damages translate directly into farm-level production costs.

To mitigate these losses, municipalities implement capture operations funded by grants from the Ministry of Agriculture, Forestry and Fisheries (MAFF). While MAFF sets the general guidelines, municipalities exercise discretion over the specific payment structure used to compensate hunters. Two primary modalities are utilized:

- **Plan A (Activity-based):** Reimbursement for inputs, primarily through per-diem allowances for hunter mobilization.
- **Plan B (Bounty-style):** Performance-based payments tied to the verified number of animals captured.

Both schemes typically rely on local hunter associations (*ryoyukai* in Japanese) for implementation, yet the nature of these contracts presents distinct incentive challenges.[2]

The institutional risk associated with **Plan A** is primarily one of moral hazard. Because compensation is decoupled from results and tied to inputs, hunters may lack the financial motivation to maximize capture efficiency. This can lead to "budget consump-

[1] For data sources and exact figures, see Appendix A.

[2] Administrative details, including specific project names and source materials, are provided in Appendix A.

tion" via per-diems without achieving the necessary threshold of removals required to alleviate agricultural pressure.

In contrast, **Plan B** provides high-powered incentives for capture volume but faces a potential "Capture-Damage Gap." This refers to a phenomenon where increased capture totals do not necessarily result in proportional damage reduction (Hone, 2007). This gap often manifests as a spatial distortion of effort: rational hunters may prioritize "easy-to-catch" animals in remote habitats—such as deep forests where deer density is high—rather than focusing on agricultural fringes where capture is more difficult but damage reduction is most vital.

Historically, deer management in Japan has struggled with this transition from simple population reduction to targeted damage mitigation (Kaji et al., 2013). Despite these theoretical concerns and the economic importance of incentive design, the relative effectiveness of these two modalities in reducing actual economic loss remains a subject of ongoing debate in wildlife policy.

2.2 Data

Data source. The main administrative dataset used in this study consists of municipality-level implementation reports submitted under Japan's *Comprehensive Grant for Wildlife Damage Prevention* program (*Chōjū Higai Bōshi Sōgō Taisaku Kōfukin*). The data were provided in electronic form by the Hokkaido Prefectural Government. I use the implementation-report data for fiscal years 2022–2024.

These implementation reports record, by municipality and program component, (i) species-specific numbers of animals captured under each plan, (ii) the number of hunters registered in each regional *ryoyūkai*, and (iii) expenditures associated with capture operations. The expenditure section reports species-specific cost items, which allows me to isolate spending related to deer management.

For Plan A, the reports provide the annual total labor cost and the total number of dispatches. Dividing total dispatches by the number of registered hunters yields the average number of dispatches per hunter in each region. For Plan B, the reports record per-head bounties, which are composed of a nationwide component and a region-specific top-up set by the municipality. I use the sum of these components as the effective per-

Table 1. Summary statistics pooled across regions and years (2022–2024)

Variable	Mean	Std	P25	P50	P75
Number of Registered Hunters	33.26	29.32	15.00	25.00	42.00
Monetary Damage (10,000 yen)	0.50	1.02	0.03	0.11	0.45
Spatial Damage	205.65	559.21	9.61	33.84	146.00
Catch (Plan A)	207,262.11	2,168,411.42	50.50	250.00	647.40
Catch (Plan B)	633.25	922.08	116.50	289.00	779.50
Per-diem Payment (yen)	1.21	0.56	0.82	1.03	1.58
Bounty (yen)	1.12	0.49	0.70	0.96	1.35

Notes. The table reports summary statistics of region-level measures pooled across years 2022–2024. The unit of observation is a region-year (447 region-years), spanning 149 regions and 3 years. Monetary damage is reported in units of 10,000 yen. For each year, variables are first aggregated within region according to the mapping described in the text/code (hunters and payments: means; damages and catches: sums), and then pooled across years. Percentiles are computed from the pooled region-year distribution. Sample sizes may vary by variable due to missing values.

head bounty. [3] Note that the data do not report dispatch activity under Plan B, as such information is not required for computing bounty payments.

Descriptive statistics. Missingness is non-trivial in the raw panel; specifically, the largest gaps are concentrated in Plan A activity/payment variables, where 290 out of 447 observations (64.9%) are missing (see Appendix Table C.1). This stems from the fact that detailed input cost decompositions are not a prerequisite for reimbursement. While we address this via neighbor-based imputation (detailed in Appendix C), we acknowledge that such augmentation is a second-best solution.

However, poor data quality is a pervasive reality in administrative datasets. The necessity of addressing real-world problems despite these limitations is precisely what motivates our focus on robustness. Since the underlying estimates cannot be viewed as absolute, the application of robust contract theory and similar frameworks becomes even more pertinent. In this sense, the "poor" quality of the data does not merely necessitate imputation—it actively justifies the use of a robust analytical approach.

Table 1 reports summary statistics for the region-level measures used in the analysis, pooled across 2022–2024. There are 149 regions in each year. The table summarizes the cross-regional distribution of key outcomes—damages and catches—as well as payments

[3] In some regions, municipalities implement a multi-tiered bounty system, e.g., higher rates for catches diverted to gibier processing. To represent the overall policy incentive in these cases, I use a weighted average of the statutory rates. This ensures the bounty reflects the effective policy intensity of the region while maintaining its character as a pre-determined administrative parameter.

and the number of registered hunters.

First, the number of registered hunters is fairly stable over time, consistent with the fact that most areas in Hokkaido experience little net population inflow and thus remain in a relatively steady demographic state. Second, damages exhibit substantial cross-regional heterogeneity: for monetary damage, the 75th percentile is about six times the 25th percentile (0.24 vs. 0.04 in units of 10,000 yen), and for spatial damage, the corresponding ratio is roughly 9.5 (95.00 vs. 10.12 hectares). Third, reflecting this heterogeneity in damages, catch outcomes also vary widely across regions under both plans.

By contrast, payment parameters display much less dispersion. For both the per-diem payment in Plan A and the per-head bounty in Plan B, the interquartile range is modest: the 75th percentile is at most about twice the 25th percentile. This limited variation suggests that municipalities may be setting payment levels with reference to nearby jurisdictions or prevailing local norms, rather than tailoring incentives tightly to local conditions. Finally, Plan B is substantially more costly than Plan A. Although the typical per-head bounty in Plan B is of the same order as the per-diem payment in Plan A, Plan B pays on every animal caught, so total outlays scale directly with catch volume.

Figure C.1 in Appendix C visualizes the regional variation in these key variables. The maps show that even adjacent regions often exhibit nontrivial differences in damages, catches, and payment levels. I exploit this within-neighborhood variation to mitigate endogeneity concerns regarding payment levels in the estimation. Details are provided in Section 4.

3 Model of Hunting

I model the number of deer hunted in region i in year t. Recall that there are two hunting schemes. Plan A is duty-based: hunters are paid a fixed daily wage $w_{i,t}$ for participating in the hunting. Plan B is incentive-based: hunters are rewarded $\tau_{i,t}$ for each deer they cull. Each regional association has a hunter base $B_{i,t}$, the number of registered hunters in the association. Over the course of the year, the association repeatedly organizes hunts under either Plan A or Plan B. The number of hunts under each plan is primarily determined by how many hunters are active. Accordingly, the mechanism behind realized hunting outcomes has two components: (1) hunters' labor supply decisions, and (2) the

production mapping labor supply into hunting results.

Consider a single hunter in region i and year t (indices omitted for brevity). He chooses the number of dispatches under Plan A and Plan B, $(d^A \geq 0, d^B \geq 0)$, and the per-dispatch effort levels under each plan, $(e^A \geq 0, e^B \geq 0)$. Let M^B denote the total number of deer hunted under Plan B by this hunter in the year. Income is

$$y = wd^A + \tau M^B,$$

where w is the per-dispatch payment under Plan A and τ is the per-head bounty under Plan B.

The hunter is risk neutral and maximizes expected net monetary payoff. The disutility of dispatches and effort is given by $c(d^A) + c(d^B)$ and $g(e^A) + g(e^B)$, with $c'(\cdot) > 0$, $c''(\cdot) \geq 0$, $g'(\cdot) > 0$, and $g''(\cdot) \geq 0$. The problem is

$$\max_{d^A, d^B, e^A, e^B} \; wd^A + \tau\, \mathbb{E}[M^B \mid e^B, d^B, x] \; - \; c(d^A) - c(d^B) \; - \; g(e^A) \; - \; g(e^B)$$

$$\text{s.t.} \quad \ln M^B \mid e^B, d^B, x \sim \mathcal{N}\big(\mu(e^B, d^B, x), \sigma_m^2\big),$$

where x summarizes the local abundance of deer. Because $\ln M^B$ is Normal with mean $\mu(e^B, d^B, x)$ and variance σ_m^2, m is lognormal and

$$\mathbb{E}[m \mid e^B, d^B, x] = \exp\big(\mu(e^B, d^B, x) + \tfrac{1}{2}\sigma_m^2\big) =: \tilde{\mu}(e^B, d^B, x).$$

The objective can therefore be written as

$$wd^A + \tau\, \tilde{\mu}(e^B, d^B, x) - c(d^A) - c(d^B) - g(e^A) - g(e^B).$$

Parametric Specifications I assume power costs for dispatches and (baseline-plus) effort:

$$c(D) = \frac{\kappa}{1+\gamma} D^{1+\gamma}, \qquad g(e) = \frac{\eta}{1+\phi} (1+e)^{1+\phi},$$

with $\kappa > 0$, $\gamma > 0$, $\eta > 0$, and $\phi > 0$.

For both plans, I specify the conditional log-mean of catch as log-linear in the abund-

ance of deer, the number of hunters, dispatches, and effort:

$$\mu(e, d, x) = \alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \alpha_d \ln d + \alpha_e \ln(1 + e).$$

Accordingly, under either Plan A or Plan B, $\ln M = \mu(e, d, x) + \xi$, where $\xi \sim \mathcal{N}(0, \sigma_m^2)$, where I assume that the log-variance σ_m^2 is common across plans and that the shocks are i.i.d. across plan realizations.

Then the conditional expected catch is

$$\begin{aligned}\mathbb{E}[m \mid e, d, x] &= \exp\Big(\alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \alpha_d \ln d + \alpha_e \ln(1 + e) + \tfrac{1}{2}\sigma_m^2\Big) \\ &= \Xi\, d^{\alpha_d}\,(1 + e)^{\alpha_e},\end{aligned}$$

where Ξ collects terms that are constant with respect to the choice variables (d, e): $\Xi \equiv \exp\big(\alpha_0 + \frac{1}{2}\sigma_m^2\big)\, N^{\alpha_N} B^{\alpha_B}$.

3.1 Solution

This section derives reduced-form expressions for the logarithm of the number of deer hunted under each scheme. Let M^A denote the number of deer hunted under Plan A, and let M^B denote the number of deer hunted under Plan B. Accordingly, I use ξ^A and ξ^B to denote the idiosyncratic shocks to log catches under each plan.

I begin by imposing a set of structural parameter restrictions that ensure the hunter's interior first-order conditions are well behaved: in particular, they admit a locally unique solution whenever an interior optimum obtains.

Assumption 1. *Assume*

$$\gamma > 0, \qquad \phi > 0, \qquad \alpha_d > 0, \qquad \alpha_e > 0, \qquad \alpha_d < 1 + \gamma, \qquad \alpha_e < 1 + \phi,$$

and

$$\Delta \equiv (\alpha_d - 1 - \gamma)(\alpha_e - 1 - \phi) - \alpha_d \alpha_e \neq 0.$$

Proposition 1 establishes reduced-form representations for the numbers of catches under Plans A and B. These expressions are obtained by solving the representative hunter's

problem for optimal dispatches and effort and then substituting the optimal choices into the log-catch equations.

Proposition 1 (Reduced forms for $\ln M^A$ and $\ln M^B$)**.** *Suppose Assumption 1 holds. Then the* interior *optimal choices implied by the hunter's first-order conditions yield*

$$\ln M^A = \alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \frac{\alpha_d}{\gamma} \ln w - \frac{\alpha_d}{\gamma} \ln \kappa + \xi^A,$$
$$\ln M^B = \psi_0 + \psi_N \ln N + \psi_B \ln B + \psi_\tau \ln \tau + \psi_\kappa \ln \kappa + \psi_\eta \ln \eta + \xi^B,$$

where the coefficients are

$$\psi_\tau = \frac{(1+\phi)\alpha_d + (1+\gamma)\alpha_e}{\Delta}, \qquad \psi_\kappa = -\frac{(1+\phi)\alpha_d}{\Delta}, \qquad \psi_\eta = -\frac{(1+\gamma)\alpha_e}{\Delta},$$

$$\psi_N = \alpha_N(1+\psi_\tau), \qquad \psi_B = \alpha_B(1+\psi_\tau),$$

and the constant term is

$$\psi_0 = \alpha_0 + (1+\psi_\tau)\Big(\alpha_0 + \tfrac{1}{2}\sigma_m^2\Big) + \frac{(1+\phi)\alpha_d}{\Delta} \ln \alpha_d + \frac{(1+\gamma)\alpha_e}{\Delta} \ln \alpha_e.$$

Proof. See Appendix B.1 □

Under Plan A, the mapping is transparent: $\frac{\partial \ln M^A}{\partial \ln w} = \frac{\alpha_d}{\gamma} > 0$ and $\frac{\partial \ln M^A}{\partial \ln \kappa} = -\frac{\alpha_d}{\gamma} < 0$. A higher per-dispatch payment increases hunting through more dispatches, while higher dispatch costs reduce hunting.

Under Plan B, the reduced form is log-linear in (τ, κ, η): in particular, Plan B catches are increasing in the bounty τ and decreasing in the cost shifters (κ, η) whenever $\Delta < 0$ (so that $\psi_\tau > 0$, $\psi_\kappa < 0$, and $\psi_\eta < 0$). This sign pattern follows the canonical incentives logic: a higher τ raises the marginal return to Plan B dispatches and effort, while higher cost shifters make dispatching and exerting effort more expensive and thus reduce activity. Empirically, we therefore expect $\psi_\tau > 0$, $\psi_\kappa < 0$, and $\psi_\eta < 0$.

4 Empirical Strategy

This section explains how I estimate the key parameters of the model developed in Section 3. A central empirical challenge is that the catch equations, specified in Proposition 1,

depend on an unobserved state: the potential abundance of deer, N. If local associations adjust the per-diem payment w or the bounty τ in response to expected abundance, then these policy variables are endogenous in the catch equations. I address this issue by exploiting cross-sectional comparisons across adjacent regions. The identifying idea is that neighboring regions are likely to face similar animal abundance in a given year. Under this approximation, taking differences between neighboring regions largely removes the latent state component, so that the remaining cross-border differences in payments can be treated as plausibly exogenous shocks.

Notation and data. There are R regions indexed by $i = 1, \ldots, R$ and periods indexed by $t = 1, \ldots, T$. Let $x_{i,t} \equiv \ln N_{i,t}$. This is the latent state variable in the following state-space model, and it is unobserved to the econometrician. For each region and period, I observe

$$y_{i,t} := \ln D_{i,t}, \qquad m^A_{i,t} := \ln M^A_{i,t}, \qquad m^B_{i,t} := \ln M^B_{i,t}, \qquad p^w_{i,t} := \ln w_{i,t}, \qquad p^\tau_{i,t} := \ln \tau_{i,t},$$

where $D_{i,t}$ is monetary damage caused by wildlife and $M^A_{i,t}$ and $M^B_{i,t}$ are the numbers of animals caught under Plans A and B, respectively. I also observe the number of hunters $B_{i,t}$ and denote its logarithm by $b_{i,t} \equiv \ln B_{i,t}$. I also utilize an adjacency matrix for all regional pairs.

Reduced-form disturbances. In the log-linear reduced forms of Proposition 1, let λ denote the reduced-form slope of Plan A log catches with respect to the log per-diem payment, $\lambda \equiv \alpha_d/\gamma$. Define the composite disturbances in the two catch equations as

$$\varepsilon^A_{i,t} \equiv -\lambda \ln \kappa_{i,t} + \xi^A_{i,t}, \qquad \varepsilon^B_{i,t} \equiv \psi_\kappa \ln \kappa_{i,t} + \psi_\eta \ln \eta_{i,t} + \xi^B_{i,t},$$

so that the reduced-form catch equations can be written compactly.

State-space model. The empirical analysis is organized around the following state-space model. First, given the policy instruments $(w_{i,t}, \tau_{i,t})$, catches under each plan are realized. Second, damage is realized conditional on the latent state and the realized catches. Lastly, the latent state transits following a transition dynamics.

1. **Beginning of period t.** The predetermined state x_t is given.

2. **Hunting stage.** A representative hunter observes the policy $(w_{i,t}, \tau_{i,t})$ and choose the effort and the number of dispatches under Plan A and Plan B. Idiosyncratic shocks, including $\kappa_{i,t}, \eta_{i,t}, \xi^A_{i,t}$ and $\xi^B_{i,t}$ are realized at this stage, and determines the number of catches in both plans $(m^A_{i,t}, m^B_{i,t})$ according to the following equations (1)–(2):

$$m^A_{i,t} = \alpha_0 + \alpha_N x_{i,t} + \alpha_B b_{i,t} + \lambda p^w_{i,t} + \varepsilon^A_{i,t}, \tag{1}$$

$$m^B_{i,t} = \psi_0 + \psi_N x_{i,t} + \psi_B b_{i,t} + \psi_\tau p^\tau_{i,t} + \varepsilon^B_{i,t}, , \tag{2}$$

   where the parameters are specified in Proposition 1.

3. **Damage observation.** Damage during period t is realized and observed as

$$y_{i,t} = \delta_i + \delta_t + \delta_N x_{i,t} + \delta^A_M m^A_{i,t} + \delta^B_M m^B_{i,t} + u_{i,t}, \tag{3}$$

   where $u_{i,t}$ is a damage innovation.

4. **State transition.** Given the predetermined state x_t and current catches $(m^A_{i,t}, m^B_{i,t})$, the end-of-period state evolves according to the following state transition equation, which allows for dependence on own states as well as on current hunting outcomes under the two plans:

$$x_{i,t+1} = \beta_i + \beta_t + \beta_N x_{i,t} + \beta^A_M m^A_{i,t} + \beta^B_M m^B_{i,t} + e_{i,t}, \tag{4}$$

   where $e_{i,t}$ is a state innovation.

4.1 Estimation of Reduced-Form Parameters

Let $\mathcal{E}$ denote the set of unordered neighboring region pairs (i, j), constructed from the adjacency matrix. For any variable $z_{i,t}$, define the border-difference transformation

$$\Delta z_{ij,t} \;\equiv\; z_{i,t} - z_{j,t}, \qquad (i,j) \in \mathcal{E},\; t = 1, \ldots, T.$$

The key maintained approximation is that adjacent regions face similar latent animal abundance in the same period, so that the border difference largely removes the unob-

served state:

$$\Delta x_{ij,t} \approx 0 \qquad \text{for } (i,j) \in \mathcal{E}.$$

I implement the estimation on the border-difference panel $\{\Delta z_{ij,t}\}_{(i,j)\in\mathcal{E},t}$.

First stage. Using (1), the border-differenced Plan A catch equation is

$$\Delta m^A_{ij,t} = \lambda\, \Delta p^w_{ij,t} + \alpha_B\, \Delta b_{ij,t} + \Gamma'_A \Delta q^A_{ij,t} + \Delta \varepsilon^A_{ij,t}, \tag{5}$$

where $\Delta q^A_{ij,t}$ collects additional differenced controls used in the empirical specification, and I omit the intercept because it cancels under differencing.

Similarly, from (2), the border-differenced Plan B catch equation is

$$\Delta m^B_{ij,t} = \psi_\tau\, \Delta p^\tau_{ij,t} + \psi_B\, \Delta b_{ij,t} + \Gamma'_B \Delta q^B_{ij,t} + \Delta \varepsilon^B_{ij,t}, \tag{6}$$

where $\Delta q^B_{ij,t}$ denotes the additional differenced controls included in the Plan B first stage.

I estimate both (5) and (6) using OLS on the border-differenced panel. In both specifications, standard errors are clustered at the region-pair level (i,j) to account for potential spatial and temporal correlations within pairs.

Second stage. The damage equation (3) motivates the border-differenced specification

$$\Delta y_{ij,t} = \delta^A_M\, \Delta m^A_{ij,t} + \delta^B_M\, \Delta m^B_{ij,t} + \delta_i - \delta_j + \Delta u_{ij,t}.$$

In practice, to absorb remaining systematic differences across borders, I estimate the second stage with region-pair fixed effects:

$$\Delta y_{ij,t} = \delta^A_M\, \Delta m^A_{ij,t} + \delta^B_M\, \Delta m^B_{ij,t} + \Pi' \Delta r_{ij,t} + \mu_{ij} + \mu_t\, \mathbf{1}\{\text{year FE}\} + \tilde{u}_{ij,t},$$

where μ_{ij} is a fixed effect for each unordered pair (i,j), $\Delta r_{ij,t}$ denotes additional differenced controls, and year fixed effects are also included.

4.2 Estimation of Structural Parameters

This subsection characterizes identification of the key structural parameters in the hunting model introduced in Section 3. The structural parameters of interest are $(\gamma, \phi, \alpha_d, \alpha_e)$, where γ governs the curvature of dispatch costs, ϕ governs the curvature of effort costs,

and (α_d, α_e) are the output elasticities with respect to dispatches and effort in the catch technology.

For the identification result, I need normalization assumptions. First, Assumption 2 fixes the scale of the latent cost shifters through variance normalizations and rules out serial correlation by imposing independence over time. Second, Assumption 3 assumes a quadratic effort cost by setting $\phi = 1$. This restriction serves as a normalization that fixes the unit scale of effort.[4]

Assumption 2 (Disturbence normalization and independence)**.** *The cost shifters* $\ln \kappa_{i,t}$ *and* $\ln \eta_{i,t}$ *are normalized to have unit variance and to be uncorrelated:*

$$\begin{aligned} &\mathbb{E}\left[\ln \kappa_{i,t}\right] = 0, \quad \mathbb{E}\left[\ln \eta_{i,t}\right] = 0, \\ &\mathrm{Var}(\ln \kappa_{i,t}) = 1, \qquad \mathrm{Var}(\ln \eta_{i,t}) = 1, \qquad \mathrm{Cov}(\ln \kappa_{i,t}, \ln \eta_{i,t}) = 0. \end{aligned}$$

In addition, the plan-specific catch shocks are independent of $(\ln \kappa_{i,t}, \ln \eta_{i,t})$*.*

All shocks are independent across equations and over time: for any $(i,t) \neq (j,s)$*,*

$$(\ln \kappa_{i,t}, \ln \eta_{i,t}, \xi^A_{i,t}, \xi^B_{i,t}, e_{i,t}, u_{i,t}) \ \perp \ (\ln \kappa_{j,s}, \ln \eta_{j,s}, \xi^A_{j,s}, \xi^B_{j,s}, e_{j,s}, u_{j,s}),$$

and, within each (i,t)*, these shocks are mutually independent.*

Assumption 3 (Effort normalization)**.** $\phi = 1$*: the effort cost takes the form* $g(e) = \frac{\eta}{2}(1+e)^2$*.*

Under Assumptions 2–3, the state-space system can be estimated by maximum likelihood given the reduced-form estimates $(\lambda, \alpha_B, \psi_\tau, \psi_B, \delta^A_M, \delta^B_M)$ obtained in Section 4.1. The MLE delivers estimates of (i) the coefficients on the latent state, (ii) the intercepts in each equation, (iii) the state transition parameters, and (iv) the second moments of the innovations.

Proposition 2 establishes point identification of the structural parameters under Assumptions 2–3. In brief, the dispatch-side parameters (γ, α_d) are pinned down from the

[4]Effort is not observed in the data and therefore has no intrinsic unit. In the model, the catch technology depends on $\ln(1+e)$, while disutility depends on $(1+e)^{1+\phi}$. Absent a normalization, a rescaling of effort can be offset by a corresponding change in the effort productivity parameter α_e and the cost shifter η without affecting the likelihood. Fixing $\phi = 1$ pins down the curvature of effort costs and anchors the scale of effort, so that α_e and η are interpretable and the remaining parameters are identified.

second moments of the reduced-form disturbances, and the effort-side elasticity α_e is then identified once the reduced-form intercept ψ_0 is identified. Accordingly, the proposition provides a set of direct estimands for the structural parameters as functions of the reduced-form coefficients and estimated second moments.

Proposition 2 (Identification)**.** *Suppose Assumption 2 and Assumption 3 hold. If* $\alpha_d > 0, \alpha_e > 0, \gamma > 0$*, then:*

1. ***Identification of*** $(\psi_\kappa, \psi_\eta, \sigma_m^2)$***.*** *If* $\lambda \neq 0$*,*

$$\psi_\kappa = -\frac{\mathrm{Cov}(\varepsilon_{i,t}^A, \varepsilon_{i,t}^B)}{\lambda}, \qquad \sigma_m^2 = \mathrm{Var}(\varepsilon_{i,t}^A) - \lambda^2,$$

and

$$\psi_\eta = -\sqrt{\mathrm{Var}(\varepsilon_{i,t}^B) - \psi_\kappa^2 - \sigma_m^2}.$$

2. ***Identification of*** γ ***and*** α_d***.*** *If* $\psi_\kappa \neq 0$ *and* $\lambda \neq 0$*,* γ *is identified by*

$$\gamma = \frac{\psi_\kappa}{\lambda\Big(\psi_\kappa\big(1 - \frac{1}{\lambda}\big) - (1 - \psi_\eta)\Big)},$$

and α_d *is identified as* $\alpha_d = \lambda\gamma$*.*

3. ***Identification of*** α_e***.*** *Suppose* $\psi_\eta \neq 0$ *and* $\psi_\kappa \neq 0$*. Then* α_e *is identified as*

$$\alpha_e = \exp\left(\frac{\alpha_0 + (1 + \psi_\tau)\big(\alpha_0 + \frac{1}{2}\sigma_m^2\big) - \psi_\kappa \ln \alpha_d - \psi_0}{\psi_\eta}\right).$$

Proof. See Appendix B.2 □

5 Estimation Result

This section presents the estimation results for both the reduced-form and structural parameters.

Reduced-form parameters. I begin with the reduced-form estimates. Each panel in Table 2 reports three quantities: (i) an OLS estimate of the coefficient attached to log catches in a regression of log damages, (ii) the first-stage relationship between payment

Table 2. Reduced-form Estimates: Plan A and Plan B

	OLS	First stage	2SLS
Panel A: Plan A			
Effect of catches on log damage ($\widehat{\delta}_M^A$)	0.005		-0.079
	(0.024)		(0.106)
Effect of payment on catches ($\widehat{\lambda}$)		0.903	
		(0.278)	
First-stage F statistic		32.136	
Panel B: Plan B			
Effect of catches on log damage ($\widehat{\delta}_M^B$)	0.131		0.086
	(0.058)		(0.417)
Effect of payment on catches ($\widehat{\psi}_\tau$)		0.471	
		(0.190)	
First-stage F statistic		30.252	
Observations	1041	1041	1041
Pairs	347	347	347

Notes. For all estimations, I use the dataset with missing values imputed according to the rule described in Appendix C. The specifications are described in Section 4.1. In the first stage, the control vector $q_{ij,t}$ includes cross-plan payment and activity controls, the number of hunters, and spatial-damage controls. In the second stage, the control vector $r_{ij,t}$ uses the same controls as $q_{ij,t}$ and additionally includes fixed effects for group-pair and year (C(group_pair_id), C(year)). OLS and first-stage standard errors are clustered by pair. 2SLS standard errors are cluster bootstrap by pair ($B = 300$).

amounts and catches, and (iii) a 2SLS estimate that uses quasi-exogenous variation in payments to identify the causal effect of catches on damages. Throughout, the second-stage specification includes group-pair fixed effects and controls for cross-plan payments. Standard errors are clustered at the pair level, and the 2SLS column reports pair-cluster bootstrap standard errors.

Plan A yields a small positive OLS coefficient, $\widehat{\delta}_M^A = 0.005$, but a negative 2SLS estimate, $\widehat{\delta}_M^A = -0.079$ (s.e. 0.106). The sign switch is consistent with upward bias in the raw OLS, while the IV estimate points to a damage-reducing effect of catches under Plan A, while the 2SLS estimate is imprecise. The first stage is clearly relevant, with $\widehat{\lambda} = 0.903$ (s.e. 0.278) and a first-stage F statistic of 32.136. Plan B shows a positive OLS association, $\widehat{\delta}_M^B = 0.131$ (s.e. 0.058), and a similarly positive but imprecise 2SLS estimate, $\widehat{\delta}_M^B = 0.086$ (s.e. 0.417). The first stage is again relevant, with $\widehat{\psi}_\tau = 0.471$ (s.e. 0.190) and $F = 30.252$.

Table 3. Structural parameter estimates

Parameter	Estimate	Std. Error	95% CI (low)	95% CI (high)
γ	0.308	0.962	-1.578	2.194
α_d	0.278	0.869	-1.425	1.981
α_e	1.931	1.148	-0.319	4.181

Notes. Structural parameters are recovered from reduced-form estimates using the mapping in Section 4.2, with normalization $\phi = 1$ fixed. The fixed normalization parameter ϕ is omitted from the table. Inference is based on FE-excluded variance propagation in the current run.

These reduced-form results first highlight how difficult it is to translate hunting activity into measurable damage reduction. In both plans, the 2SLS estimates are statistically imprecise, suggesting that neither regime delivers a precisely estimated catch-to-damage effect in this specification. Still, the point estimates are informative: Plan A implies a negative causal effect, whereas Plan B does not. This pattern suggests that the Plan A scheme may be more effective to damage mitigation than Plan B, even if uncertainty remains substantial. The contrast is also consistent with a potentially noisy mapping from measured catches to realized damages under outcome-contingent incentives (Kaji et al., 2013). Under Plan B, measured catches may reflect margins that are weakly linked to local damage reduction, such as reporting or spatial reallocation margins, which can attenuate the estimated effect on damages. Based on this evidence, the subsequent simulation exercises use the Plan A estimate as the reference point for the catch-to-damage link.

Structural parameters. Table 3 reports the structural parameter estimates. Under the normalization $\phi = 1$, the remaining parameters $(\gamma, \alpha_d, \alpha_e)$ are recovered from the reduced-form moments using the mapping in Section 4.2. Inference is obtained by the delta method.

Quantitatively, the estimates are $\hat{\gamma} = 0.308$, $\hat{\alpha}_d = 0.278$, and $\hat{\alpha}_e = 1.931$, with the normalization $\phi = 1$. A key implication is an explicit asymmetry between dispatch and effort. On the cost side, effort is more costly at the margin than dispatch because $\phi > \hat{\gamma}$, i.e., the effort cost schedule is more convex. On the productivity side, however, effort is also far more effective for generating catches, since $\hat{\alpha}_e \gg \hat{\alpha}_d$. This asymmetry creates a non-trivial policy trade-off. Inducing effort is attractive because it is the more productive margin, but it is also the margin with steeper marginal cost. Inducing dispatch is cheaper

at the margin, but it contributes less to catches per unit behavioral response. Therefore, the policy question is not mechanically "induce effort" or "induce dispatch"; rather, it is an empirical question.

At the same time, statistical precision remains limited: standard errors are large and the 95% confidence intervals include zero (Table 3). Hence, I treat these estimates as reference point on the relative roles of dispatch and effort, and place greater weight on robustness exercises described in Section 6.

State-space model parameters. Table C.4 in Appendix C reports the estimated parameters of the state-space model—the catch equations, the damage equation, and the state-transition equation—excluding parameters that are mechanically fixed or plugged in from earlier reduced-form steps.

Several patterns emerge. First, persistence in the latent state is tightly estimated: $\hat{\beta}_N = 1.008$ (s.e. 0.107; 95% CI $[0.798, 1.218]$), indicating strong intertemporal persistence in latent deer abundance. By contrast, the direct effects of current catches on next period's state, β_{MA} and β_{MB}, are very imprecisely estimated, with wide confidence intervals that include zero. Thus, the data do not provide sharp evidence that current management measurably shifts next-period abundance. A plausible interpretation is that observed damage reduction, when present, may operate mainly through contemporaneous removal of damage-causing deer rather than through a clearly identifiable stock-reduction effect in the next period.

Second, in the catch equations, the latent-state loadings are close to zero and imprecise. This suggests that, conditional on controls and fixed effects, short-run variation in measured catches is driven more by policy/payment variation and idiosyncratic shocks than by cross-sectional differences in the latent state.

Third, the shock estimates indicate substantial residual noise in hunting outcomes and state dynamics: $\sigma_m = 1.467$ (s.e. 0.977), $\sigma_u = 1.422$ (s.e. 0.714), and $\sigma_e = 0.200$ (s.e. 6.324). Among these, only σ_u has a confidence interval that excludes zero.

These features directly motivate the robust-contract analysis. Specifically, because (i) abundance is persistent, (ii) dynamic catch effects on next-period abundance are weakly identified, and (iii) outcome equations are noisy, a policy calibrated to a single point estimate is fragile. In response to these results, the robust design in Section 6 prioritizes payment rules that deliver stable damage reduction under model uncertainty rather than

maximizing performance only at the estimated mean.

6 Contract Design

This section evaluates alternative contracts using the estimates above. Specifically, taking the filtered state at the end of the sample period as given, I compute a set of contracts for that period and compare their predicted performance with observed outcomes, including catches and damages.

I consider two classes of contract design. The baseline is the *optimal contract*, defined as the contract that minimizes expected damage subject to a budget constraint under the estimated model.

The second is a *robust contract* that explicitly accounts for model ambiguity. In this exercise, I allow ambiguity in: (i) the effectiveness of catches in reducing damages, parameterized by δ_M^A and δ_M^B in the empirical model; and (ii) the structural parameters governing hunters' choices, including γ (the curvature of dispatch costs) and (α_d, α_e) (the productivity of dispatch and effort in producing catches).

The first source of ambiguity reflects hard-to-model behavioral margins, especially under bounty incentives. When payments are tied to measured catches, opportunistic reporting is a recurring concern. Bounty incentives may also reallocate effort toward "easy-to-hunt" remote areas in ways that do not reduce—and may even increase—damages in residential or agricultural zones through spatial displacement (Hone, 2007).

The second source of ambiguity reflects estimation imprecision. Since the data do not pin down structural parameters with high precision, it is natural for the policymaker to account for parameter uncertainty directly in contract design. For example, the true dispatch-cost curvature may be higher than the point estimate, which would materially change the cost-effectiveness of dispatch-based incentives.

Specifically, I define the robust contract as a *max–min robust* contract: among feasible contracts, the policymaker chooses the one that delivers the best performance in the worst-case model within a prespecified ambiguity set. This notion of robustness is standard in the robust-contract literature as in Carroll (2015). I adopt max–min robustness because it better matches the institutional environment of public decision-making. In administrative settings, decisions are typically evaluated under conservative accountability standards, where avoiding severe downside outcomes is a primary objective. A worst-case

performance criterion is therefore a natural benchmark.[5]

6.1 Setting

Fix a region i and period t. Let the latent state be $x_{i,t} \equiv \ln N_{i,t}$ and the observed log-damage be $y_{i,t} \equiv \ln D_{i,t}$, as in Section 4. At the beginning of period t, the policymaker observes the filtered state $\hat{x}_{i,t|t}$.[6] The policymaker then announces a *contract*, specified below. Hunters determine the number of dispatch and effort level through period t, a total number of deer are harvested, damage is realized.

Let $M_{i,t}$ denote the number of deer harvested in region i during period t: so the per-hunter number of harvested deer is $\frac{M_{i,t}}{B_{i,t}}$ where $B_{i,t}$ is the number of registered hunters. And let $d_{i,t}$ denote the number of dispatches chosen by the representative hunter. In other words, total dispatches in a region is $d_{i,t} \times B_{i,t}$. As in the practical contracts, both $\frac{M_{i,t}}{B_{i,t}}$ and $d_{i,t}$ are contractible, in the sense that the contract may condition the monetary transfer on these variables. The policymaker's objective is not catches per se, but the monetary damage caused by deer. We do *not* treat monetary damage as contractible.

A region- and time-specific hunting contract is a payment schedule $s_{i,t} : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$ that specifies the transfer to the representative hunter in region i during period t as a function of the realized number of dispatches $d_{i,t}$ and the realized per-hunter catch $\frac{M_{i,t}}{B_{i,t}}$. Hence, the total transfer to the region is computed as $s_{i,t} \times B_{i,t}$ Let $\mathcal{D}_{i,t} \subseteq \mathbb{R}_+$ denote a policymaker-specified feasible set for dispatches.[7] Hereafter, when no confusion is likely, I also use M to denote per-hunter catch.

Administrative constraints. Let $\mathcal{S}_{i,t}^{\text{adm}}$ be a policymaker-specified class of contracts encoding institutional restrictions. In the empirical application, the class is specified by the following three constraints.

The first one is feasibility which restricts the contract space into the natural one. A

[5] Other robustness criteria are also available, such as regret minimizing contract as in Bergemann and Schlag (2008). Regret-minimizing robust contracts are also computationally feasible in this framework and provide a useful direction for future applications.

[6] This inference about the latent state is obtained via the Kalman filter.

[7] Imposing an explicit upper bound on dispatches rules out pathological contracts whose incentives are defined only off the equilibrium path. In applications, $\mathcal{D}_{i,t}$ can be taken as a compact interval $[0, \bar{d}_{i,t}]$, where $\bar{d}_{i,t}$ is chosen large enough to contain all empirically relevant dispatch levels.

contract $s_{i,t}$ is feasible if it belongs to the set

$$\mathcal{S}_{i,t}^{\text{fea}} \equiv \left\{ \begin{array}{l} s : \mathcal{D}_{i,t} \times \mathbb{R}_+ \to \mathbb{R}_+ \ \text{ s.t.} \\ s(d, M) \geq 0 \quad \forall (d, M) \in \mathcal{D}_{i,t} \times \mathbb{R}_+, \\ s(\cdot, M) \text{ is nondecreasing on } \mathcal{D}_{i,t} \quad \forall M \geq 0, \\ s(d, \cdot) \text{ is nondecreasing on } \mathbb{R}_+ \quad \forall d \in \mathcal{D}_{i,t}. \end{array} \right\}.$$

This feasibility requirement imposes minimal institutional realism on the payment schedule. Nonnegativity rules out punitive transfers from hunters to the authority. Monotonicity in dispatches requires that, holding catches fixed, a hunter who undertakes more dispatch activity is not paid less. Monotonicity in catches requires that, holding dispatches fixed, a hunter who delivers more measured output is not penalized.

The second restriction imposes *separability with linearity.* Contracts are separable in dispatches and catches, and I further restrict each component to a simple parametric form:

$$\mathcal{S}_{i,t}^{\text{sep-lin}} \equiv \Big\{ s : \mathcal{D}_{i,t} \times \mathbb{R}_+ \to \mathbb{R}_+ \ \Big| \ \exists\, (w, b) \in \mathbb{R}_+ \times \mathbb{R}_+ \text{ such that} \\ s(d, M) = t(d) + q(M),\ t(d) = w\, d,\ q(M) = b\, M \Big\}.$$

The dispatch component is linear in dispatches, with per-dispatch rate w, and the catch component is also linear, with slope b. I impose $w \geq 0$ and $b \geq 0$, so that neither dispatch activity nor measured catches is penalized. This parsimonious contract class is close to the payment rules used in current practice, which makes the resulting policy implications more directly implementable.[8]

The third restriction is the per-period budget constraint. For a contract s, let $\mathbb{P}_{\hat{\theta}}^{(d,M)}(\cdot \mid x, s)$ denote the contract-induced joint distribution of dispatches and catches defined below. The expected-payment constraint is

$$\mathcal{S}_{i,t}^{\text{bud}} \equiv \Big\{ s : \mathcal{D}_{i,t} \times \mathbb{R}_+ \to \mathbb{R}_+ \ \Big| \ \mathbb{E}_{\mathbb{P}_{\hat{\theta}}^{(d,M)}(\cdot|x,s)}\big[s(d, M)\big] \ \leq \ \bar{S}_{i,t} \Big\},$$

where $\bar{S}_{i,t}$ is the policymaker's per-period budget for transfers in region i at time t, which

[8]The framework can be extended to richer schedules such as truncated-linear or threshold contracts under which bounty payments apply only above a minimum catch level. Such extensions are straightforward to compute. However, even if these designs improve model-based outcomes, they typically require larger administrative changes and higher implementation capacity, so their practical feasibility in the near term may be limited.

is set to the actually total payment observed at the period in my data.

Putting these together, the administrative class is

$$\mathcal{S}_{i,t}^{\text{adm}} \equiv \mathcal{S}_{i,t}^{\text{fea}} \;\cap\; \mathcal{S}_{i,t}^{\text{sep-lin}} \;\cap\; \mathcal{S}_{i,t}^{\text{bud}}.$$

Damage realization. Fix (x, s) and consider the representative hunter in Section 3. He chooses dispatches $d \geq 0$ and per-dispatch effort $e \geq 0$ to maximize expected net payoff:

$$\max_{d,e\geq 0} \; \mathbb{E}[s(d, M) \mid d, e, x] \;-\; \frac{\kappa}{1+\gamma} d^{1+\gamma} \;-\; \frac{\eta}{1+\phi}(1+e)^{1+\phi},$$

subject to the catch technology

$$\begin{aligned} m \mid d, e, x &\sim \mathcal{N}\Big(\mu(d, e, x),\, \sigma_m^2\Big), \\ \mu(d, e, x) &= \alpha_0 + \alpha_N x + \alpha_B \ln B + \alpha_d \ln d + \alpha_e \ln(1+e). \end{aligned} \tag{7}$$

Let $(d^\star(s, x), e^\star(s, x))$ denote an optimal choice. Evaluating (7) at $(d^\star, e^\star)$ yields the induced conditional law of catches $\mathbb{P}_{\hat{\theta}}^{M}(\cdot \mid x, s)$, and hence the induced conditional law of the pair (d, M), denoted $\mathbb{P}_{\hat{\theta}}^{(d,M)}(\cdot \mid x, s)$.[9]

As in Section 4, the log-damage equation is linear in the log-catches: denoting the effectiveness of catches on damage reduction by δ_M, the damage equation is as follows,

$$y = \widehat{\delta_i} + \widehat{\delta_t} + \widehat{\delta_N}\hat{x}_{i,t|t} + \delta_M\, m + u,$$

where the other parameters in the equation is set to the estimates values.

Ambiguity set. I allow model ambiguity along four dimensions: (i) the catch-to-damage linkage, δ_M; (ii) effort productivity in catches, α_e; (iii) dispatch-cost curvature, γ; and (iv) dispatch productivity in catches, α_d. Let $\hat{\theta}$ denote the baseline estimate from Section 5, and let $\hat{\theta}^{-(\delta_M, \alpha_e, \gamma, \alpha_d)}$ collect all remaining parameters fixed at their baseline values.

[9] Formally, $\mathbb{P}_{\hat{\theta}}^{(d,M)}(\cdot \mid x, s)$ is the pushforward measure obtained by combining the (possibly degenerate) choice $d^\star(s, x)$ with the conditional distribution of M given $(d^\star, e^\star, x)$ under (7).

I construct a rectangular ambiguity set

$$\mathcal{A} = [\underline{\delta}_M, \overline{\delta}_M] \times [\underline{\alpha}_e, \overline{\alpha}_e] \times [\underline{\gamma}, \overline{\gamma}] \times [\underline{\alpha}_d, \overline{\alpha}_d],$$

where each interval is generated by multiplicative scaling around the corresponding baseline estimate. In the implementation, I use a common scaling range (0.2, 5.0), i.e., each ambiguous parameter is allowed to vary from 20% to 500% of its baseline value.

Formally, for each contract s and state x, outcomes are evaluated under the conditional law

$$\mathbb{P}_{(\hat{\theta}^{-(\delta_M,\alpha_e,\gamma,\alpha_d)},\, \delta_M,\alpha_e,\gamma,\alpha_d)}(\cdot \mid x, s),$$

which is induced by the contract-dependent catch block and the damage equation with $(\delta_M, \alpha_e, \gamma, \alpha_d)$ substituted as above.

Contract design problem. Given filtered state $x_{i,t|t}$, the optimal contract solves

$$\min_{s \in \mathcal{S}_{i,t}^{\text{adm}}} \; \mathbb{E}_{\mathbb{P}_{(\hat{\theta})}}\left[D \;\middle|\; x_{i,t|t},\; s\right], \tag{8}$$

where $\hat{\theta}$ is the baseline parameter vector from Section 5. The robust design solves the corresponding max–min problem over the 4-dimensional ambiguity set:

$$\min_{s \in \mathcal{S}_{i,t}^{\text{adm}}} \; \sup_{(\delta_M,\alpha_e,\gamma,\alpha_d) \in \mathcal{A}} \; \mathbb{E}_{\mathbb{P}_{(\hat{\theta}^{-(\delta_M,\alpha_e,\gamma,\alpha_d)},\, \delta_M,\, \alpha_e,\, \gamma,\, \alpha_d)}}\left[D \;\middle|\; x_{i,t|t},\; s\right].$$

6.2 Result

Optimal contract. I begin by describing the form of the optimal contract. Solving (8) for each region, I find that $b^\star = 0$ and $w^\star > 0$ in all regions. This implies that the optimal contract does not use a bounty payment per catch, but instead relies on a per-diem payment for each dispatch. The optimal per-diem rate $w^\star$ is typically higher than the observed per-diem payment in Plan A. Overall, the optimal contract uses higher per-diem compensation to induce more dispatches and does not employ bounty-based incentives to intensify hunting effort.

As discussed in Section 5, increasing dispatch is less productive for damage reduction, but its marginal cost is milder than that of effort. The optimal contract pattern therefore suggests that, in the current empirical application, the cost-side advantage of dispatch

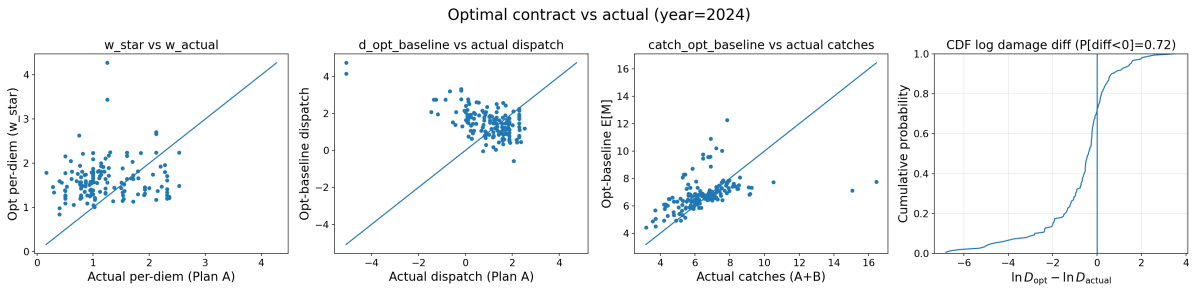

Figure 1. Optimal contract performance: per-diem pay, dispatches, catches, and damages.

Notes. Panel (1) compares the implied per-diem payment $w^\star$ under the designed contract with the observed per-diem payment in Plan A. Panel (2) compares the model-implied expected dispatches d with the observed average dispatches per hunter in Plan A. Panel (3) compares the model-implied expected catches $\mathbb{E}[M]$ with observed total catches (Plan A + Plan B). Panel (4) plots the empirical CDF of $\ln D_{\text{opt}} - \ln D_{\text{actual}}$; negative values indicate that the designed contract implies lower predicted damages than those observed in the data, where D_{opt} is the expected damage under the optimal contract. The monetary unit scale is JPY 10,000.

dominates the productivity-side advantage of effort.

Figure 1 compares model-implied outcomes under the optimal contract with observed outcomes in the data for the last sample period. The first panel plots the optimal per-diem rate, $w^\star$, against the observed per-diem payment in Plan A. The second panel compares the model-implied expected number of dispatches under the optimal contract with the observed number of Plan A dispatches. For each region, the model-implied dispatch level is computed from the hunters' best response given $(w^\star, b^\star)$. The third panel compares model-implied expected catches, $\mathbb{E}[M]$, under the optimal contract with observed total catches (Plan A plus Plan B). Both variables in the second and the third panel are shown on a log scale. The fourth panel reports the empirical CDF of the log-damage difference, $\ln D_{\text{opt}} - \ln D_{\text{actual}}$, where D_{opt} is model-implied damage under the optimal contract and D_{actual} is realized damage in the data. Values below zero indicate that model-implied damage under the optimal contract is lower than realized damage, whereas values above zero indicate the opposite.

The first panel indicates that the optimal per-diem payment is compressed into a relatively narrow range and is not uniformly above the observed Plan A payment. In low-payment regions, $w^\star$ is often higher than observed; in high-payment regions, $w^\star$ is frequently lower. This pattern suggests partial reallocation rather than a uniform upward shift in per-diem rates.

The second panel compares model-implied dispatches with observed Plan A dispatches. The cloud is centered above the 45-degree line for many observations, indicating

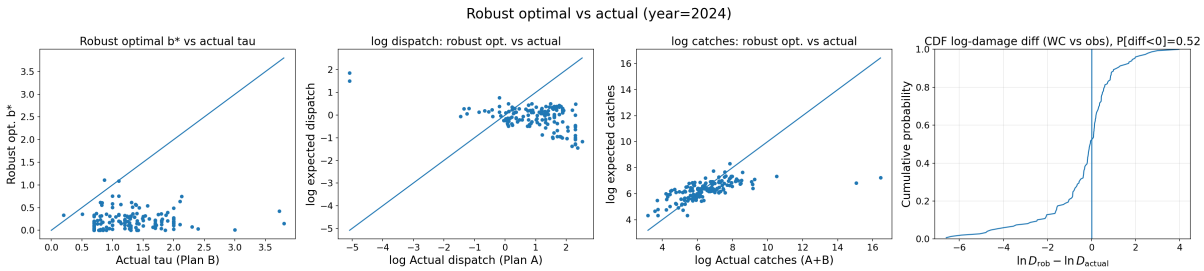

Figure 2. Robust contract performance: bounty, dispatches, catches, and damages.
Notes. Panel (1) compares the implied bounty $b^\star$ under the robust contract with the observed bounty in Plan B. Panel (2) compares the model-implied expected dispatches d under the robust contract with the observed average dispatches per hunter in Plan A. Panel (3) compares the model-implied expected catches $\mathbb{E}[M]$ under the robust contract with observed total catches (Plan A + Plan B). Panel (4) plots the empirical CDF of $\ln D_{\text{rob}} - \ln D_{\text{actual}}$; negative values indicate that the robust contract implies lower predicted damages than those observed in the data, where D_{rob} is expected damage under the robust contract. The monetary unit scale is JPY 10,000.

that the optimal contract tends to raise dispatches in regions with low observed dispatch activity, while not expanding dispatches one-for-one everywhere. In other words, the contract mainly equalizes dispatch intensity across regions rather than scaling it proportionally. This is consistent with the first panel.

In the third panel, most observations lie near the 45-degree line, with a mild upward tilt around the center of the distribution. This indicates moderate catch gains under the optimal contract for many regions, but not uniformly large increases; a few high-catch regions remain below the line.

In the fourth panel, the estimated mass to the left of zero is about 0.72, implying that in roughly 72% of regions the model-implied damage under the optimal contract is lower than realized damage. While the realized damages include shocks and measurement noise, the distribution is shifted toward negative values, consistent with meaningful damage-reduction potential from the optimal per-diem-centered design.

Robust contract. I next compute the robust contract. In contras to the optimal contract, the robust contract chooses $b^\star > 0$ and set all the per-diem payment to 0. This pattern is consistent with a robust-control interpretation of the two instruments. A per-diem payment mainly loads on the dispatch margin, so its performance is exposed to misspecification in that single channel. By contrast, a bounty payment loads on catches and therefore operates through both dispatch and effort responses. As a result, even if one behavioral path turns out to be weak in the worst case, the other path can still transmit incentives. In this sense, the bounty instrument is a hedge across margins: it may not

maximize average-case performance, but it can deliver better downside protection under joint ambiguity in $(\delta_M, \alpha_e, \gamma, \alpha_d)$.

This rationale for the robust-contract shape also helps explain why current policy uses two payment schemes in parallel. When there is uncertainty about how hunters behaviorally respond to incentives, policymakers may value protection against worst-case outcomes in wildlife-damage reduction. Because bounty-based payments in Plan B load on both dispatch and effort margins, they can still generate behavioral responses even if one margin (e.g., dispatch) is less effective than expected.

Figure 2 shows the performance of the robust contract. The first panel compares $b^\star > 0$ computed as a robust contract with the observed bounty in Plan B. Most points lie below the 45-degree line, indicating that the robustly chosen bounty slope is typically lower than the observed Plan B bounty level.

The second panel shows that model-implied dispatches under the robust contract are generally below observed Plan A dispatches. This is consistent with the contract shape: because the robust design places weight on bounty rather than per-diem payments, it does not directly subsidize dispatch frequency.

The third panel indicates that model-implied expected catches are of similar order to observed total catches, but are often somewhat lower, especially in the upper tail. Thus, compared with observed outcomes, the robust contract does not seek aggressive expansion of hunting activity.

The fourth panel reports the CDF of $\ln D_{\text{rob}} - \ln D_{\text{actual}}$. The mass to the left of zero is about 0.52, so the robust contract predicts lower damages than observed in roughly half of regions. This is weaker than the optimal benchmark, but it is aligned with the role of robust design: it sacrifices some average-case gains to reduce sensitivity to adverse parameter realizations. In short, the robust contract behaves as a conservative, downside-protection rule rather than an average-performance maximizer.

The policy implication is twofold. First, bounty-based incentives can be justified as a robustness device: they hedge against model uncertainty by activating multiple behavioral margins and thereby improving worst-case protection. Second, the current bounty level appears excessive when the policy goal is minimizing wildlife damage. In the robust-contract results, the implied bounty slope is typically concentrated around roughly one-half to one-quarter of the observed bounty of Plan B, yet it delivers dispatch and catch outcomes that are broadly comparable to those realized in the data. This

suggests that the current schedule may be operating in a high-cost, low-marginal-gain region of hunter utility. A practical reform is therefore not to remove bounty payments, but to recalibrate them downward while preserving their robustness role.

7 Conclusion

This paper studies wildlife-damage control as a contract-design problem for procuring hunting activity under uncertainty. Using a structural state-space framework with administrative data from Hokkaido, I show that policy conclusions depend on the decision criterion. Under a baseline objective, the model selects a per-diem-centered corner contract ($w > 0,\ b = 0$). Under a max–min robust objective, however, the solution flips to a bounty-centered corner ($w = 0,\ b > 0$), reflecting the value of hedging across behavioral margins when key parameters are ambiguous. The main policy implication is therefore not a binary choice between per-diem-based and bounty-based schemes, but a disciplined recalibration of payment levels and instrument mix: preserve bounty incentives as a robustness device, while reducing overly high bounty rates and using dispatch-based payments where average-case efficiency is well supported by the data.

Beyond wildlife management, the choice between cost-plus and fixed-price contracts is central to procurement and regulation (Lafontaine and Slade, 2012). By linking structural estimation to contract optimization, the framework proposed here offers a template for evidence-based policy design in other sectors. However, general application faces the challenge of endogeneity: compensation terms often correlate with unobserved task difficulty. While this study utilized spatial variation to isolate incentives from environmental confounders, future work in other domains must similarly prioritize credible identification strategies to disentangle the causal effects of incentives.

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A Institutional and Data Details

This appendix provides the technical, administrative, and data sources supporting the context described in Section 2.

Administrative Framework and Legal Basis The administrative capture of wildlife for damage control is legally grounded in the *Wildlife Protection and Management and Hunting Management Act* (Article 9).[10] Financial support is primarily channeled through the Ministry of Agriculture, Forestry and Fisheries (MAFF) under the "Grant for Comprehensive Measures for Wildlife Damage Prevention" (*Chojuhigai Boshi Sogo Taisaku Kofukin*). This grant operates based on municipal damage-prevention plans and is governed by specific implementation guidelines.[11]

Payment Modalities (Plan A and Plan B) The MAFF grant framework allows municipalities to select between two distinct project types, which form the basis of the payment modalities analyzed in this paper:

- **Plan A (Comprehensive Support Project):** This corresponds to the *Comprehensive Support Project* component of the grant. Budgets are formulated primarily in terms of activity-based line items—such as per-diem allowances, operating expenses, and contracted services—rather than explicit pay-for-output. Implementing entities (municipalities) prepare an annual plan, and funds can be disbursed via advance payment or reimbursement. Settlement is based on performance reports filed at the fiscal year-end, focusing on the execution of the planned activities rather than a strict per-head count.

- **Plan B (Emergency Capture Activity Support Project):** This corresponds to the *Emergency Capture Activity Support Project*. It functions as a bounty-style system where payments are proportional to the number of animals captured. Crucially, the per-head bounty is conditional on verified capture within the fiscal year (April 1 to March 31). Operationally, payment is triggered by strict documentation. The guideline mandates a standardized capture-confirmation form recording

[10] The Act is available via e-Gov: `https://laws.e-gov.go.jp/law/414AC0000000088/`.

[11] MAFF Implementation Guideline (*Jisshi Yoryo*): `https://www.maff.go.jp/j/seisan/tyozyu/higai/yosan/attach/pdf/yosan-187.pdf`. Disbursement/Settlement Rules (*Kofu-to Yoko*): `https://www.maff.go.jp/j/seisan/tyozyu/higai/yosan/attach/pdf/yosan-175.pdf`. A summary is available at: `https://www.maff.go.jp/j/seisan/tyozyu/higai/yosan/attach/pdf/yosan-164.pdf`.

the capturer's identity, species, location, and method of confirmation. Practical verification often requires attaching photographic evidence to auditable records.

Implementing Actors (Ryoyukai) Capture operations are typically executed by local hunter associations, known as *ryoyukai*. The system is organized as a nationwide network with prefecture-level associations and local branches.[12] While the Act does not legally restrict permits to association members, MAFF operational manuals emphasize that nuisance-control teams are often best organized through local *ryoyukai* branches, establishing them as the de facto infrastructure for implementation.[13]

Data on Damage (Hokkaido) The economic significance of deer in Hokkaido is supported by prefectural statistics. In FY2022, the Hokkaido Government reported total wildlife-related damages to agriculture, forestry, and fisheries of 5,887 million JPY. Of this, 4,846 million JPY (approximately 82%) was attributed to Ezo deer.[14]

Damages are heavily concentrated in forage crops, which are critical for Hokkaido's dairy industry. Pasture and forage grass damages accounted for 1,795 million JPY, representing about 37% of all deer-attributed losses. This concentration underscores the direct link between deer management and the cost structure of the region's livestock sector.[15]

[12]For the organizational structure, see *Dainihon Ryoyukai*: `https://j-hunters.com/about/organization.php`. An example of a prefectural chapter (*Miyagi-ken Ryoyukai*): `https://miyagi-ryoyukai.com/`.

[13]MAFF wildlife-damage prevention manual (case materials): `https://www.maff.go.jp/j/seisan/tyozyu/higai/manyuaru/attach/pdf/8_old_manual-3.pdf`.

[14]Source: Hokkaido Government, "Yasei Choju ni yoru Norinsuisangyo Higai Chosa Kekka (Reiwa 4 nendo)" [Survey Results on Wildlife Damage to Agriculture, Forestry, and Fisheries (FY2022)]: `https://www.pref.hokkaido.lg.jp/ks/skn/higai.html`. Dollar conversion assumes 153.32 JPY/USD (Bank of Japan spot rate, Jan 29, 2026): `https://www.boj.or.jp/statistics/market/forex/fxdaily/fxlist/fx260129.pdf`.

[15]For an overview of Hokkaido's agriculture, see NARO: `https://www.naro.go.jp/english/laboratory/harc/overview/index.html`. For background on forage production: `https://ocw.hokudai.ac.jp/wp-content/uploads/2016/01/AgricultureInHokkaido-2009-Text-04.pdf`.

B Omitted Proofs

B.1 Proof of Proposition 1

Fix a region-year and suppress indices. Recall the dispatch and effort costs

$$c(d) = \frac{\kappa}{1+\gamma} d^{1+\gamma}, \qquad g(e) = \frac{\eta}{1+\phi}(1+e)^{1+\phi},$$

and the Plan B expected catch technology

$$\mathbb{E}[m \mid e^B, d^B, x] = \Xi\,(d^B)^{\alpha_d}(1+e^B)^{\alpha_e}, \qquad \Xi := \exp\!\big(\alpha_0 + \tfrac{1}{2}\sigma_m^2\big)\, N^{\alpha_N} B^{\alpha_B}.$$

The hunter chooses (d^A, d^B, e^A, e^B) to maximize

$$w d^A + \tau\,\Xi\,(d^B)^{\alpha_d}(1+e^B)^{\alpha_e} - c(d^A) - c(d^B) - g(e^A) - g(e^B).$$

Plan A. Since effort does not affect Plan A income in this formulation, $e^{A\star} = 0$. The first-order condition for d^A is

$$w = c'(d^{A\star}) = \kappa (d^{A\star})^{\gamma} \quad \Rightarrow \quad d^{A\star} = \left(\frac{w}{\kappa}\right)^{1/\gamma}.$$

Therefore,

$$\begin{aligned} \ln M^A &= \alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \alpha_d \ln d^{A\star} + \alpha_e \ln(1+e^{A\star}) \\ &= \alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \frac{\alpha_d}{\gamma}\big(\ln w - \ln \kappa\big) + \xi^A, \end{aligned}$$

which yields the first reduced form in Proposition 1.

Plan B. The first-order condition for e^B is

$$\tau \Xi \alpha_e (d^{B\star})^{\alpha_d}(1+e^{B\star})^{\alpha_e - 1} = \eta (1+e^{B\star})^{\phi}.$$

The first-order condition for d^B equates the marginal benefit of Plan B dispatches to the marginal dispatch cost under Plan B:

$$\tau \Xi \alpha_d (d^{B\star})^{\alpha_d - 1}(1+e^{B\star})^{\alpha_e} = c'(d^{B\star}) = \kappa (d^{B\star})^{\gamma}.$$

Let $u := \ln d^{B\star}$ and $v := \ln(1 + e^{B\star})$. Taking logs of these two conditions yields

$$(\alpha_d - 1 - \gamma)u + \alpha_e v = \ln\kappa - \ln\tau - \ln\Xi - \ln\alpha_d,$$
$$\alpha_d u + (\alpha_e - 1 - \phi)v = \ln\eta - \ln\tau - \ln\Xi - \ln\alpha_e.$$

Let

$$A := \ln\kappa - \ln\tau - \ln\Xi - \ln\alpha_d, \qquad B := \ln\eta - \ln\tau - \ln\Xi - \ln\alpha_e,$$

and define

$$\Delta \equiv (\alpha_d - 1 - \gamma)(\alpha_e - 1 - \phi) - \alpha_d\alpha_e.$$

If $\Delta \neq 0$, solving the 2×2 system implies the composite term

$$\begin{aligned}\alpha_d u + \alpha_e v &= \frac{1}{\Delta}\Big\{ - \alpha_d(1+\phi)A - \alpha_e(1+\gamma)B\Big\} \\ &= \frac{(1+\phi)\alpha_d + (1+\gamma)\alpha_e}{\Delta}\ln\tau - \frac{(1+\phi)\alpha_d}{\Delta}\ln\kappa - \frac{(1+\gamma)\alpha_e}{\Delta}\ln\eta \\ &\quad + \frac{(1+\phi)\alpha_d + (1+\gamma)\alpha_e}{\Delta}\ln\Xi + \frac{(1+\phi)\alpha_d}{\Delta}\ln\alpha_d + \frac{(1+\gamma)\alpha_e}{\Delta}\ln\alpha_e.\end{aligned}$$

Using

$$\ln\Xi = \alpha_0 + \tfrac{1}{2}\sigma_m^2 + \alpha_N \ln N + \alpha_B \ln B,$$

and the definition

$$\ln M^B = \alpha_0 + \alpha_N \ln N + \alpha_B \ln B + \alpha_d u + \alpha_e v + \xi^B,$$

we obtain

$$\ln M^B = \psi_0 + \psi_N \ln N + \psi_B \ln B + \psi_\tau \ln\tau + \psi_\kappa \ln\kappa + \psi_\eta \ln\eta + \xi^B,$$

where

$$\psi_\tau = \frac{(1+\phi)\alpha_d + (1+\gamma)\alpha_e}{\Delta}, \qquad \psi_\kappa = -\frac{(1+\phi)\alpha_d}{\Delta}, \qquad \psi_\eta = -\frac{(1+\gamma)\alpha_e}{\Delta},$$

$$\psi_N = \alpha_N(1+\psi_\tau), \qquad \psi_B = \alpha_B(1+\psi_\tau),$$

and

$$\psi_0 = \alpha_0 + (1+\psi_\tau)\Big(\alpha_0 + \tfrac{1}{2}\sigma_m^2\Big) + \frac{(1+\phi)\alpha_d}{\Delta}\ln\alpha_d + \frac{(1+\gamma)\alpha_e}{\Delta}\ln\alpha_e.$$

This proves Proposition 1. □

B.2 Proof of Proposition 2

Throughout, fix (i, t) and suppress indices.

Step 1: Reduced-form disturbances. By construction, the reduced-form disturbances in the two catch equations are

$$\varepsilon^A = -\lambda \ln \kappa + \xi^A, \qquad \varepsilon^B = \psi_\kappa \ln \kappa + \psi_\eta \ln \eta + \xi^B,$$

where $\lambda \equiv \alpha_d/\gamma$. Under Assumption 2, $\ln \kappa$ and $\ln \eta$ have unit variances and are uncorrelated, and ξ^A and ξ^B are mean-zero, mutually independent, and independent of $(\ln \kappa, \ln \eta)$, with

$$\xi^A \sim \mathcal{N}(0, \sigma_m^2), \qquad \xi^B \sim \mathcal{N}(0, \sigma_m^2), \qquad \mathrm{Cov}(\xi^A, \xi^B) = 0.$$

Step 2: Identification of ψ_κ. Using mutual independence and $\mathrm{Cov}(\xi^A, \ln \kappa) = \mathrm{Cov}(\xi^A, \ln \eta) = 0$,

$$\begin{aligned}
\mathrm{Cov}(\varepsilon^A, \varepsilon^B) &= \mathrm{Cov}\Big(-\lambda \ln \kappa + \xi^A,\ \psi_\kappa \ln \kappa + \psi_\eta \ln \eta + \xi^B \Big) \\
&= -\lambda \psi_\kappa \,\mathrm{Var}(\ln \kappa) - \lambda \psi_\eta \,\mathrm{Cov}(\ln \kappa, \ln \eta) + \mathrm{Cov}(\xi^A, \xi^B) \\
&= -\lambda \psi_\kappa,
\end{aligned}$$

where we used $\mathrm{Var}(\ln \kappa) = 1$, $\mathrm{Cov}(\ln \kappa, \ln \eta) = 0$, and $\mathrm{Cov}(\xi^A, \xi^B) = 0$. Therefore, if $\lambda \neq 0$,

$$\psi_\kappa = -\frac{\mathrm{Cov}(\varepsilon^A, \varepsilon^B)}{\lambda}.$$

Step 3: Identification of σ_m^2 and ψ_η. First,

$$\begin{aligned}
\mathrm{Var}(\varepsilon^A) &= \mathrm{Var}\big(-\lambda \ln \kappa + \xi^A \big) \\
&= \lambda^2 \,\mathrm{Var}(\ln \kappa) + \mathrm{Var}(\xi^A) + 2\,\mathrm{Cov}(-\lambda \ln \kappa, \xi^A) \\
&= \lambda^2 + \sigma_m^2,
\end{aligned}$$

so

$$\sigma_m^2 = \mathrm{Var}(\varepsilon^A) - \lambda^2.$$

Next,

$$\begin{aligned}
\mathrm{Var}(\varepsilon^B) &= \mathrm{Var}\big(\psi_\kappa \ln\kappa + \psi_\eta \ln\eta + \xi^B\big) \\
&= \psi_\kappa^2 \,\mathrm{Var}(\ln\kappa) + \psi_\eta^2 \,\mathrm{Var}(\ln\eta) + \mathrm{Var}(\xi^B) + 2\psi_\kappa\psi_\eta \,\mathrm{Cov}(\ln\kappa, \ln\eta) \\
&\quad + 2\,\mathrm{Cov}(\psi_\kappa \ln\kappa + \psi_\eta \ln\eta, \xi^B) \\
&= \psi_\kappa^2 + \psi_\eta^2 + \sigma_m^2,
\end{aligned}$$

where we used $\mathrm{Var}(\ln\kappa) = \mathrm{Var}(\ln\eta) = 1$, $\mathrm{Cov}(\ln\kappa, \ln\eta) = 0$, and independence of ξ^B from $(\ln\kappa, \ln\eta)$. Therefore,

$$|\psi_\eta| = \sqrt{\mathrm{Var}(\varepsilon^B) - \psi_\kappa^2 - \sigma_m^2}.$$

Under the maintained sign convention (equivalently, under $\Delta < 0$), we take

$$\psi_\eta = -\sqrt{\mathrm{Var}(\varepsilon^B) - \psi_\kappa^2 - \sigma_m^2},$$

which establishes part (i).

Step 4: Point identification of γ and α_d. From Plan A we have $\lambda = \alpha_d/\gamma$, hence $\alpha_d = \lambda\gamma$.

Next, recall from the Plan B reduced form derived from the separated-cost model that

$$\psi_\kappa = -\frac{(1+\phi)\alpha_d}{\Delta}, \qquad \psi_\eta = -\frac{(1+\gamma)\alpha_e}{\Delta},$$

and that

$$\Delta = (\alpha_d - 1 - \gamma)(\alpha_e - 1 - \phi) - \alpha_d\alpha_e.$$

Eliminating (α_e, ϕ) yields a compatibility restriction linking (γ, α_d) to (ψ_κ, ψ_η). Substitute $\alpha_d = \lambda\gamma$ and rearrange the defining relation for Δ using the two expressions above; the common factor $(1+\phi)$ cancels, and we obtain the scalar equation

$$1 = \psi_\eta + \frac{\alpha_d - 1 - \gamma}{\alpha_d}\,\psi_\kappa = \psi_\eta + \frac{\lambda\gamma - 1 - \gamma}{\lambda\gamma}\,\psi_\kappa. \tag{9}$$

If $\psi_\kappa \neq 0$ and $\lambda \neq 0$, (9) is linear in $1/\gamma$, and solving gives

$$\gamma = \frac{\psi_\kappa}{\lambda\Big(\psi_\kappa\big(1-\frac{1}{\lambda}\big) - (1-\psi_\eta)\Big)},$$

Therefore γ and $\alpha_d = \lambda\gamma$ are point identified, proving part (ii).

Step 5: Point identification of (α_e, ϕ). Finally, we show that the effort-related parameters are point identified once the reduced-form intercept ψ_0 is identified. Using the two Plan B coefficient relations above,

$$\psi_\kappa = -\frac{(1+\phi)\alpha_d}{\Delta}, \qquad \psi_\eta = -\frac{(1+\gamma)\alpha_e}{\Delta},$$

we obtain

$$\frac{\psi_\kappa}{\psi_\eta} = \frac{(1+\phi)\alpha_d}{(1+\gamma)\alpha_e}, \qquad \text{so that} \qquad 1+\phi = \frac{(1+\gamma)\alpha_e}{\alpha_d}\cdot\frac{\psi_\kappa}{\psi_\eta}.$$

To pin down α_e, we use the reduced-form intercept relation for Plan B. Under the normalization $\mathbb{E}[\ln\kappa] = \mathbb{E}[\ln\eta] = 0$, the intercept satisfies

$$\psi_0 = \alpha_0 + (1+\psi_\tau)\Big(\alpha_0 + \tfrac{1}{2}\sigma_m^2\Big) + \frac{(1+\phi)\alpha_d}{\Delta}\ln\alpha_d + \frac{(1+\gamma)\alpha_e}{\Delta}\ln\alpha_e.$$

Substituting $\frac{(1+\phi)\alpha_d}{\Delta} = -\psi_\kappa$ and $\frac{(1+\gamma)\alpha_e}{\Delta} = -\psi_\eta$ gives

$$\psi_0 = \alpha_0 + (1+\psi_\tau)\Big(\alpha_0 + \tfrac{1}{2}\sigma_m^2\Big) - \psi_\kappa\ln\alpha_d - \psi_\eta\ln\alpha_e,$$

and hence

$$\ln\alpha_e = \frac{\alpha_0 + (1+\psi_\tau)\big(\alpha_0 + \frac{1}{2}\sigma_m^2\big) - \psi_\kappa\ln\alpha_d - \psi_0}{\psi_\eta}.$$

Therefore,

$$\alpha_e = \exp\left(\frac{\alpha_0 + (1+\psi_\tau)\big(\alpha_0 + \frac{1}{2}\sigma_m^2\big) - \psi_\kappa\ln\alpha_d - \psi_0}{\psi_\eta}\right).$$

Plugging this value into (B.2) yields a unique value of ϕ,

$$\phi = \frac{(1+\gamma)\alpha_e}{\alpha_d}\cdot\frac{\psi_\kappa}{\psi_\eta} - 1.$$

Thus (α_e, ϕ) is point identified, establishing part (iii). □

B.3 Likelihood

This appendix derives the likelihood function for the state-space model in Section 4. I first rewrite the model in stacked matrix form, then impose Gaussianity, and finally state the Kalman-filter prediction-error likelihood.

Stacked vector notation Let $x_t \in \mathbb{R}^R$ collect the latent states across regions,

$$x_t := (x_{1,t}, \ldots, x_{R,t})',$$

and let $m_t^A, m_t^B, y_t, p_t^w, p_t^\tau, b_t \in \mathbb{R}^R$ stack the corresponding observables:

$$m_t^A := (m_{1,t}^A, \ldots, m_{R,t}^A)', \quad m_t^B := (m_{1,t}^B, \ldots, m_{R,t}^B)', \quad y_t := (y_{1,t}, \ldots, y_{R,t})',$$

$$p_t^w := (p_{1,t}^w, \ldots, p_{R,t}^w)', \quad p_t^\tau := (p_{1,t}^\tau, \ldots, p_{R,t}^\tau)', \quad b_t := (b_{1,t}, \ldots, b_{R,t})'.$$

Let $W \in \mathbb{R}^{R\times R}$ denote the (row-normalized) spatial-weight matrix with (i,j) element w_{ij} and $\mathbf{1}_R$ the $R \times 1$ vector of ones. Let I_R be the $R \times R$ identity matrix.

For the policy equations, stack the instruments as $Z_t \in \mathbb{R}^{R\times K}$ (where the ith row is $Z_{i,t}'$), and define the associated $R \times 1$ vectors

$$\nu_t^w := (\nu_{1,t}^w, \ldots, \nu_{R,t}^w)', \qquad \nu_t^\tau := (\nu_{1,t}^\tau, \ldots, \nu_{R,t}^\tau)'.$$

Similarly define $e_t := (e_{1,t}, \ldots, e_{R,t})'$ and $u_t := (u_{1,t}, \ldots, u_{R,t})'$. For the catch disturbances, define $\varepsilon_t^A := (\varepsilon_{1,t}^A, \ldots, \varepsilon_{R,t}^A)'$ and $\varepsilon_t^B := (\varepsilon_{1,t}^B, \ldots, \varepsilon_{R,t}^B)'$.

Policy equations The policy stage (**??**)–(**??**) can be written as

$$\begin{aligned} p_t^w &= \omega^w + \omega_t^w + \pi_B^w\, b_t + Z_t \Pi^w + \nu_t^w, \\ p_t^\tau &= \omega^\tau + \omega_t^\tau + \pi_B^\tau\, b_t + Z_t \Pi^\tau + \nu_t^\tau, \end{aligned} \tag{10}$$

where $\omega^w, \omega^\tau \in \mathbb{R}^R$ collect region fixed effects and $\omega_t^w, \omega_t^\tau \in \mathbb{R}^R$ collect year fixed effects (equation-specific). The vectors $\Pi^w, \Pi^\tau \in \mathbb{R}^K$ are the instrument coefficients.

Catch equations (measurement equations) The hunting-stage equations (1)–(2) become

$$m_t^A = \alpha_0 \mathbf{1}_R + \alpha_N x_t + \alpha_B b_t + \lambda\, p_t^w + \varepsilon_t^A,$$
$$m_t^B = \psi_0 \mathbf{1}_R + \psi_N x_t + \psi_B b_t + \psi_\tau\, p_t^\tau + \varepsilon_t^B.$$

State transition Stacking (4) across regions yields the transition equation

$$x_t = \beta + \beta_t + \Big(\beta_N I_R + \beta_S W\Big) x_{t-1} + \beta_M^A m_t^A + \beta_M^B m_t^B + e_t, \tag{11}$$

where $\beta, \beta_t \in \mathbb{R}^R$ collect region and year fixed effects and β_M^A, β_M^B multiply m_t^A, m_t^B elementwise (i.e., they are scalars applied to the $R \times 1$ vectors).

Damage measurement The damage equation (3) can be written as

$$y_t = \delta + \delta_t + \delta_N x_t + \delta_S W x_t + \delta_M^A m_t^A + \delta_M^B m_t^B + u_t, \tag{12}$$

where $\delta, \delta_t \in \mathbb{R}^R$ are region and year fixed effects.

Linear-Gaussian state-space form For likelihood construction, collect the observables in

$$Y_t := \begin{pmatrix} m_t^A \\ m_t^B \\ y_t \\ p_t^w \\ p_t^\tau \end{pmatrix} \in \mathbb{R}^{5R}.$$

Let C_t denote the collection of observed controls (including b_t and Z_t) and fixed effects. Then (10)–(12) imply a linear measurement representation of the form

$$Y_t = c_t(C_t) + H x_t + \xi_t, \tag{13}$$

where $c_t(C_t) \in \mathbb{R}^{5R}$ stacks the deterministic components and

$$H := \begin{pmatrix} \alpha_N I_R \\ \psi_N I_R \\ \delta_N I_R + \delta_S W \\ 0_{R\times R} \\ 0_{R\times R} \end{pmatrix} \in \mathbb{R}^{5R\times R}, \qquad \xi_t := \begin{pmatrix} \varepsilon_t^A \\ \varepsilon_t^B \\ u_t \\ \nu_t^w \\ \nu_t^\tau \end{pmatrix}.$$

The state transition (11) can be written as

$$x_t = a_t(C_t) + Fx_{t-1} + G\zeta_t, \tag{14}$$

where

$$F := \beta_N I_R + \beta_S W,$$

and the transition shock is $\zeta_t := e_t$ with $G := I_R$. The vector $a_t(C_t) \in \mathbb{R}^R$ stacks the fixed effects and the terms involving m_t^A, m_t^B (which are included in Y_t and hence are known at time t when forming the conditional density).

Gaussian disturbances Assume the disturbances are Gaussian and satisfy Assumption 2. In particular, for each (i,t),

$$\ln \kappa_{i,t} \sim \mathcal{N}(0,1), \qquad \ln \eta_{i,t} \sim \mathcal{N}(0,1), \qquad \mathrm{Cov}(\ln \kappa_{i,t}, \ln \eta_{i,t}) = 0,$$

and all shocks are mutually independent across equations and over time. Since

$$\varepsilon_{i,t}^A = -\lambda \ln \kappa_{i,t}, \qquad \varepsilon_{i,t}^B = \psi_\kappa \ln \kappa_{i,t} + \psi_\eta \ln \eta_{i,t},$$

the stacked measurement disturbance ξ_t is mean zero Gaussian with covariance matrix $R_t := \mathrm{Var}(\xi_t)$ that is block diagonal across regions and has the within-region block

$$R^{(i)} := \mathrm{Var}\begin{pmatrix} \varepsilon_{i,t}^A \\ \varepsilon_{i,t}^B \\ u_{i,t} \\ \nu_{i,t}^w \\ \nu_{i,t}^\tau \end{pmatrix} = \begin{pmatrix} \lambda^2 & -\lambda\psi_\kappa & 0 & 0 & 0 \\ -\lambda\psi_\kappa & \psi_\kappa^2 + \psi_\eta^2 & 0 & 0 & 0 \\ 0 & 0 & \mathrm{Var}(u_{i,t}) & 0 & 0 \\ 0 & 0 & 0 & \mathrm{Var}(\nu_{i,t}^w) & 0 \\ 0 & 0 & 0 & 0 & \mathrm{Var}(\nu_{i,t}^\tau) \end{pmatrix},$$

where $\mathrm{Var}(u_{i,t})$ and $\mathrm{Var}(\nu_{i,t}^{\cdot})$ are parameters (or normalized, if desired). Likewise, the transition disturbance $\zeta_t = e_t$ is Gaussian with covariance

$$Q_t := \mathrm{Var}(\zeta_t) = \mathrm{Var}(e_t).$$

Under Assumption 2, these covariances are time-invariant and independent across regions, so $R_t \equiv R$ and $Q_t \equiv Q$.

Likelihood via prediction errors Let $\mathcal{F}_{t-1}$ denote the sigma-field generated by the data up to $t-1$. Given the linear-Gaussian system (13)–(14), the Kalman filter yields one-step-ahead predictions

$$\hat{x}_{t|t-1} := \mathbb{E}[x_t \mid \mathcal{F}_{t-1}], \qquad P_{t|t-1} := \mathrm{Var}(x_t \mid \mathcal{F}_{t-1}),$$

and the implied prediction error for the observables,

$$v_t := Y_t - \hat{Y}_{t|t-1}, \qquad \hat{Y}_{t|t-1} := c_t(C_t) + H\hat{x}_{t|t-1}.$$

The prediction-error covariance is

$$S_t := \mathrm{Var}(v_t \mid \mathcal{F}_{t-1}) = HP_{t|t-1}H' + R.$$

Under Gaussianity, the conditional density is

$$Y_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\hat{Y}_{t|t-1}, S_t),$$

so the log-likelihood contribution at time t is

$$\ell_t(\theta) = -\frac{1}{2}\Big(5R\ln(2\pi) + \ln|S_t| + v_t' S_t^{-1} v_t\Big),$$

where θ collects all parameters in the state-space model. Summing over t gives the sample log-likelihood,

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \ell_t(\theta).$$

The maximum likelihood estimator is

$$\hat{\theta} \in \arg\max_{\theta} \ \mathcal{L}(\theta).$$

Initialization. To run the Kalman filter, one specifies an initial condition $x_0 \sim \mathcal{N}(\hat{x}_{0|0}, P_{0|0})$. In practice, I either treat $(\hat{x}_{0|0}, P_{0|a0})$ as additional parameters or use a diffuse prior (initialization with a large variance), depending on the specification.

C Additional Figures and Tables

Table C.1. Progress of Missing-Value Imputation by Variable

Variable	Miss. Init.	Step A	Step B	Step C	Miss. Final	Fill Rate (%)
Dispatch Count (Plan A)	290	0	26	264	0	100.0
Per-diem Payment (Plan A)	290	0	26	264	0	100.0
Total Cost (Plan A)	290	0	26	264	0	100.0
Total Hunters (Plan A)	290	0	26	264	0	100.0
Catch (Plan A)	249	0	120	129	0	100.0
Num. Hunter	180	0	54	126	0	100.0
Spatial Damage (Target)	111	87	0	24	0	100.0
Spatial Damage (Realized)	110	0	89	21	0	100.0
Spatial Damage (Past)	108	87	0	21	0	100.0
Monetary Damage (Realized)	95	0	83	12	0	100.0
Catch (Plan B)	92	0	56	36	0	100.0
Pay per Catch (Plan B)	92	0	56	36	0	100.0
Total Payment (Plan B)	92	0	56	36	0	100.0
Monetary Damage (Past)	91	79	0	12	0	100.0
Monetary Damage (Target)	90	78	0	12	0	100.0

Notes: Step A propagates region-level baseline/target values within region across years. Step B applies within-region time interpolation. Step C applies neighbor-based spatial imputation (iterative synchronous updates) with fallback rules. "Missing (Initial)" is computed after panel completion (region × year) and variable-specific missing conversion.

Missing-value imputation procedure. This table summarizes the staged imputation algorithm used to construct the analysis variables. Let r index regions and t years. I first form a complete (r, t) panel by taking the Cartesian product of all regions and all years in the sample, and then merging the raw data onto this grid. For selected outcome/payment variables, observed zeros are treated as missing in the preprocessing step.

Imputation is then implemented in three stages. **Step A** applies to baseline/target variables and propagates region-level values within each region across years so that values observed for a region in any year can be used for that region's missing years. This process is not restrictive because the target and the base values are not altered but fixed for all the three years in my data period. **Step B** applies within each region over time, filling missing values by time interpolation when neighboring years are observed. **Step C** performs geography-based imputation year by year: for each missing (r, t), I replace it with the mean of currently available values in adjacent regions (shared-border adjacency), iterating synchronously up to a fixed maximum number of iterations. "Synchronously" means updates in a given iteration are computed from the previous iteration's vector, avoiding order dependence.

If missing entries remain after neighbor iterations (e.g., isolated components or no valid adjacent observations), I use sequential fallback rules: (i) the same-year cross-region mean, (ii) the full-sample mean for that variable, and (iii) zero only as a last resort. The table reports, for each variable, the number of missing observations before imputation, the number filled at each step (A–C), and the final number remaining.

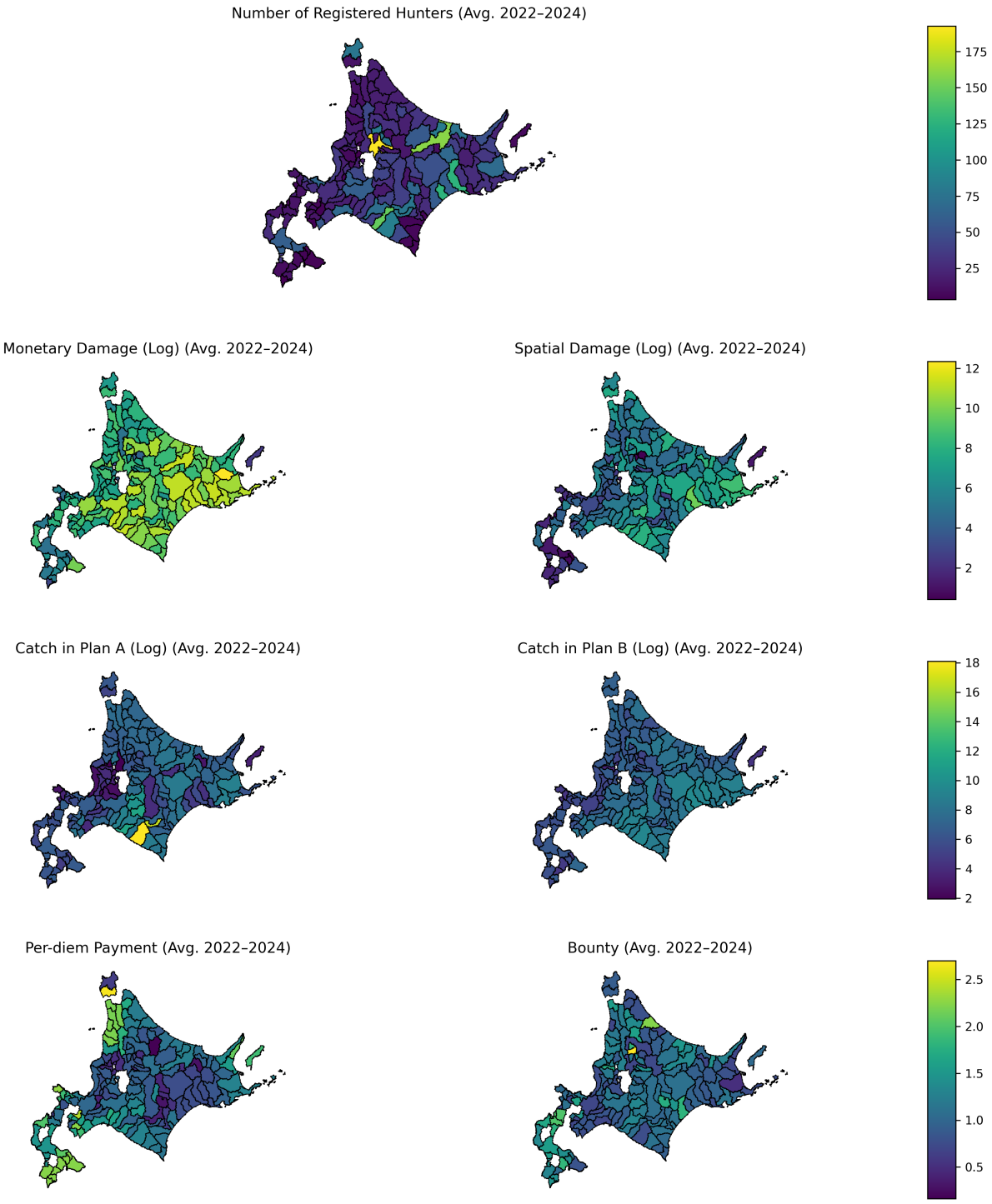

Figure C.1. Regional distribution of key variables pooled across 2022–2024.

Notes. Each panel reports a region-level measure aggregated over years 2022–2024. For each region, the number of registered hunters and payment variables are averaged over years, while damage and catch variables are summed over years and then transformed as $\log(1+x)$. Color scales are shared within each row (hunters; damages; catches; payments). Areas with missing data are shown in gray with hatching.

Table C.2. Balance checks for Set2 IVs under full controls

Panel A. Mean-difference tests

Variable	Mean(Δx)	SE	p-value	N
Num. Hunter	-1.559	7.210	0.829	1041
Monetary Damage (Past)	-393.391	3249.324	0.904	1041
Spacial Damage (Past)	4353.151	4748.974	0.359	1041
Monetary Damage (Target)	-262.924	2454.493	0.915	1041
Spacial Damage (Target)	-293.448	195.064	0.132	1041

Panel B-A. Partial balance aligned to Plan A first stage

B-A1. Associations with ΔZ_{money} and ΔZ_{space}

Variable	ΔZ_{money} coef (SE)	p	ΔZ_{space} coef (SE)	p	$F_{Z,\text{joint}}$	N
Num. Hunter	-0.293 (1.59)	0.853	1.211 (0.71)	0.088	1.47	1041
Monetary Damage (Past)	171743.236 (96851.75)	0.076	124050.797 (87890.50)	0.158	1.76	1041
Spacial Damage (Past)	-3351.031 (5256.74)	0.524	6423.321 (4012.03)	0.109	2.25	1041

B-A2. Payment term and joint exclusion test

Variable	Δw_A coef (SE)	p	F_{all}	N
Num. Hunter	-0.335 (3.96)	0.933	0.99	1041
Monetary Damage (Past)	120815.186 (80172.85)	0.132	1.17	1041
Spacial Damage (Past)	3536.481 (3578.77)	0.323	1.51	1041

B-A3. Reference only: policy-design variables

Variable	$F_{Z,\text{joint}}$	F_{all}	N
Monetary Damage (Target)	4.18	4.21	1041
Spacial Damage (Target)	4.03	4.81	1041

Panel C-B. Partial balance aligned to Plan B first stage

C-B1. Associations with ΔZ_{money} and ΔZ_{space}

Variable	ΔZ_{money} coef (SE)	p	ΔZ_{space} coef (SE)	p	$F_{Z,\text{joint}}$	N
Num. Hunter	0.357 (1.47)	0.807	1.175 (0.75)	0.119	1.26	1041
Monetary Damage (Past)	177425.372 (98720.49)	0.072	121679.112 (86772.45)	0.161	1.77	1041
Spacial Damage (Past)	-2767.893 (5307.15)	0.602	6196.265 (3958.77)	0.118	2.08	1041

C-B2. Payment term and joint exclusion test

Variable	Δp_{tB} coef (SE)	p	F_{all}	N
Num. Hunter	-3.238 (4.91)	0.510	0.93	1041
Monetary Damage (Past)	-94799.570 (85487.20)	0.267	1.18	1041
Spacial Damage (Past)	-4972.694 (3828.74)	0.194	1.41	1041

C-B3. Reference only: policy-design variables

Variable	$F_{Z,\text{joint}}$	F_{all}	N
Monetary Damage (Target)	0.56	1.48	1041
Spacial Damage (Target)	3.89	2.68	1041

Notes. All regressions use pair-difference outcomes built from neighboring-region pairs. Fixed effects include C(group_pair_id) (year FE excluded in this run). Standard errors are clustered by pair. Panel A reports mean-difference tests: $\Delta x \sim 1$+FE. Panels B-A and C-B report partial-balance regressions aligned to first-stage control sets.

Table C.3. RD/IV Estimates: Plan A vs. Plan B (Non-neighbor; No FE; Log RR Instruments)

	OLS	First stage	2SLS
Panel A: Plan A (duty-based)			
Effect of catches on log damage ($\widehat{\delta}_M^A$)	-0.016		-0.343
	(0.045)		(0.277)
Effect of payment on catches ($\widehat{\lambda}$)		1.030	
		(1.926)	
First-stage F statistic (payment only)		0.284	
First-stage F statistic (joint excluded: payment + log RR)		2.208	
Panel B: Plan B (bounty-based)			
Effect of catches on log damage ($\widehat{\delta}_M^B$)	0.276		-3.037
	(0.602)		(3.631)
Effect of payment on catches ($\widehat{\psi}_\tau$)		2.163	
		(1.019)	
First-stage F statistic (payment only)		9.560	
First-stage F statistic (joint excluded: payment + log RR)		3.211	
Observations	37	37	37
Pairs	26	26	26

Notes. Each panel reports estimates from a single specification using non-neighbor (raw) variables. The OLS and second-stage (2SLS) regressions are estimated without fixed effects. The first stage includes the payment instrument (Plan A: $\Delta \log w_A$; Plan B: $\Delta \log p_{tB}$) and two additional excluded instruments given by the pair-differences in log reduction-rate targets, $\Delta Z^{\log}_{\text{money}} = \Delta \log(\text{target/base})$ and $\Delta Z^{\log}_{\text{space}} = \Delta \log(\text{target/base})$, constructed from `Z_money_log_target_base` and `Z_space_log_target_base`. First-stage controls follow the printed formulas in the output (including the other plan's catches/payment and additional controls). Standard errors in parentheses. OLS and first-stage standard errors are clustered by pair. 2SLS standard errors are obtained by cluster bootstrap by pair ($B = 300$).

Table C.4. Parameter Estimates of State-Space Model

Parameter	Estimate	Std. Error	95% CI (low)	95% CI (high)
α_0	3.579	3.661	-3.595	10.754
ψ_0	3.865	3.176	-2.360	10.090
α_N	-0.00161	0.004125	-0.009690	0.006477
ψ_N	-0.00136	0.003328	-0.007879	0.005165
δ_N	-0.00679	0.006248	-0.019	0.005460
β_N	1.008	0.107	0.798	1.218
β_{MA}	0.134	7.031	-13.646	13.914
β_{MB}	0.181	13.099	-25.492	25.854
ψ_κ	1.194	4.400	-7.430	9.818
ψ_η	5.423	29.657	-52.704	63.550
σ_m	1.467	0.977	-0.448	3.382
σ_u	1.422	0.714	0.023	2.821
σ_e	0.200	6.324	-12.195	12.595

Notes. This table reports reduced-form estimates inferred from the state-space/regression system. Inference is based on FE-excluded variance propagation in the current run.